

# TMPRSS2 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes TMPRSS2 gene fusions from the msk\_impact\_50k\_2026 study, covering 867 fusion events. 260/867 fusions (30.0%) were in-frame. 53/867 fusions (6.1%) retained the Trypsin domain. Domain retention was tested with Fisher's exact test ( $p=0.0292045$ , statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for TMPRSS2.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 867 analyzed fusion events, identifying 73 distinct fusion partner genes. The most frequent partner was ERG, observed in 764 events.

### Domain retention

Trypsin domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 23/260 (8.8%) of in-frame fusions retained the domain,  $p=0.0292045$  (statistically significant at  $\alpha=0.05$ ). The Trypsin appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

### Domain disruption

Disruption of the Scavenger receptor cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 237/260 (91.2%) of in-frame fusions disrupted the domain,  $p=0.969277$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Cutpoint detection

Cutpoint detection scanned 849 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 88 aa (permutation-corrected  $p=0.019802$ , statistically significant at  $\alpha=0.05$ ). This is -102 aa from the nearest configured domain boundary at 190 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=53$ ) and "not\_retained" ( $n=796$ ). For Frame\_status, retained: 23/46 (50.0%, 95% CI 36.1%-63.9%); not\_retained: 237/734 (32.3%, 95% CI 29.0%-35.8%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 38.3962 vs 53.2226,  $p=0.0150037$  (statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score produced results for this run (73 row(s) in composite\_evidence\_ranking); see the accompanying table.

## Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: TMPRSS2 has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| ERG          | 764         | 0.881199538638985     | 0.0                    | 0.20043213737826426     | 0.8226056165307499       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.5105809812757611  | 1    |
| BRCA2        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.9827089337175793       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2705826380233357  | 2    |
| AIRE         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.9164265129682997       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26064027491094377 | 3    |
| MRPS6        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8904899135446686       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2567497849973991  | 4    |
| ACP3         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 5    |
| ARF4         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 6    |
| ARHGAP26     | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 7    |
| BACH1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 8    |
| C2CD2        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 9    |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| CASZ1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 10   |
| DYRK1A       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 11   |
| ELAPOR1      | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 12   |
| NSUN4        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 13   |
| PAXBP1       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 14   |
| POLDIP3      | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 15   |
| SKIL         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 16   |
| SLC60A1      | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 17   |
| TSPAN4       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8703170028818443       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25372384839797546 | 18   |
| NR3C2        | 2           | 0.002306805074971165  | 0.0                    | 0.20043213737826426     | 0.8674351585014409       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25363759250216067 | 19   |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| FOXP1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8674351585014409       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.253291571740915   | 20   |
| PALS1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8674351585014409       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.253291571740915   | 21   |
| PGM5         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8674351585014409       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.253291571740915   | 22   |
| SETD4        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8674351585014409       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.253291571740915   | 23   |
| ZNF827       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8674351585014409       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.253291571740915   | 24   |
| ETV1         | 14          | 0.016147635524798153  | 0.0                    | 0.20043213737826426     | 0.8258542610127624       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25155270701380694 | 25   |
| ETV5         | 4           | 0.00461361014994233   | 0.0                    | 0.20043213737826426     | 0.8357348703170029       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2495745907969863  | 26   |
| BRAF         | 4           | 0.00461361014994233   | 0.0                    | 0.20043213737826426     | 0.8242074927953891       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24784548416874422 | 27   |
| CHD3         | 3           | 0.0034602076124567475 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24404125015101438 | 28   |
| RIPK4        | 3           | 0.0034602076124567475 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24404125015101438 | 29   |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| MAD1L1       | 2           | 0.002306805074971165  | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24369522938976873 | 30   |
| MX1          | 2           | 0.002306805074971165  | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24369522938976873 | 31   |
| NLGN1        | 2           | 0.002306805074971165  | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24369522938976873 | 32   |
| ABCC4        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 33   |
| ADHFE1       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 34   |
| AFDN         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 35   |
| ARRB1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 36   |
| CCN6         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 37   |
| CERS6        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 38   |
| CHAF1B       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 39   |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| CHST12       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 40   |
| CSK          | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 41   |
| CYP3A43      | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 42   |
| CYP4Z1       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 43   |
| DENND3       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 44   |
| ENPP6        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 45   |
| FIRRM        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 46   |
| GCOM1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 47   |
| LSS          | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 48   |
| MYL5         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 49   |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| OSBPL1A      | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 50   |
| OSBPL2       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 51   |
| RBPMS2       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 52   |
| RCN1         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 53   |
| SEPTIN11     | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 54   |
| SH3BGR       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 55   |
| SH3GL3       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 56   |
| SIK1         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 57   |
| SIMC1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 58   |
| U2AF1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.8011527377521614       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24334920862852305 | 59   |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| DSCAM        | 3           | 0.0034602076124567475 | 0.0                    | 0.20043213737826426     | 0.7790585975024016       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2407271291135504  | 60   |
| PRDM15       | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.7723342939481268       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.23902644205791784 | 61   |
| PREX2        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.659942363112392        | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.22216765243255762 | 62   |
| ABCG1        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.48991354466858794      | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19666332966598704 | 63   |
| MGA          | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.48991354466858794      | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19666332966598704 | 64   |
| ETV4         | 3           | 0.0034602076124567475 | 0.0                    | 0.20043213737826426     | 0.261287223823247        | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.16306142306167723 | 65   |
| KLK3         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.2305475504322767       | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.15775843053054034 | 66   |
| KRTAP10-4    | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.21902017291066278      | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.15602932390229826 | 67   |
| PLCD3        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.21613832853025938      | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.15559704724523776 | 68   |
| DMD          | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | None                     | 0.827438497288003          | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.14491329172435158 | 69   |
| DYM          | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | None                     | 0.827438497288003          | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.14491329172435158 | 70   |

| Partner_gene | Event_count | Recurrence_score      | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|-----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| RSL24D1      | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | None                     | 0.827438497288003          | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.14491329172435158 | 71   |
| TUT7         | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | None                     | 0.827438497288003          | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.14491329172435158 | 72   |
| SGMS2        | 1           | 0.0011534025374855825 | 0.0                    | 0.20043213737826426     | 0.0                      | 0.827438497288003          | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.12317629796569884 | 73   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value               | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|-----------------------|---------------------|
| 19       | 27                     | 26               | 521                    | 275              | 0.5481322899749004  | 0.037858671743290474  | 1.421834627157186   |
| 38       | 27                     | 26               | 522                    | 274              | 0.5450928381962865  | 0.03737895940426984   | 1.427372793146433   |
| 42       | 34                     | 19               | 641                    | 155              | 0.4327120453239182  | 0.007549763421125502  | 2.122066657180019   |
| 43       | 46                     | 7                | 776                    | 20               | 0.16936671575846834 | 0.0008335597161986539 | 3.0790632818748924  |
| 50       | 46                     | 7                | 777                    | 19               | 0.1606913035484464  | 0.0006485489316794361 | 3.188057251720709   |
| 88       | 46                     | 7                | 779                    | 17               | 0.14340729873464148 | 0.0003764704147316645 | 3.4242691475473035  |
| 94       | 47                     | 6                | 779                    | 17               | 0.1709456568249893  | 0.0019738608138091126 | 2.7046834747291166  |
| 99       | 47                     | 6                | 780                    | 16               | 0.1606837606837607  | 0.0015320085607629643 | 2.814738807887665   |
| 101      | 47                     | 6                | 781                    | 15               | 0.15044814340588988 | 0.0011700038506478247 | 2.9318127089270436  |
| 102      | 47                     | 6                | 783                    | 13               | 0.13005534269902086 | 0.0006451030583527046 | 3.1903708991653907  |
| 117      | 51                     | 2                | 783                    | 13               | 0.42337164750957856 | 0.2395219092329614    | 0.6206547552189791  |
| 146      | 52                     | 1                | 783                    | 13               | 0.8633461047254151  | 0.5973372189026775    | 0.22378042437820417 |
| 167      | 52                     | 1                | 784                    | 12               | 0.7959183673469388  | 0.5700816283558604    | 0.24406295449180584 |
| 186      | 52                     | 1                | 785                    | 11               | 0.7286624203821656  | 0.541018269048284     | 0.2667880694352085  |
| 206      | 53                     | 0                | 785                    | 11               | None                | 1.0                   | -0.0                |
| 228      | 53                     | 0                | 786                    | 10               | None                | 1.0                   | -0.0                |
| 265      | 53                     | 0                | 788                    | 8                | None                | 1.0                   | -0.0                |
| 337      | 53                     | 0                | 789                    | 7                | None                | 1.0                   | -0.0                |
| 355      | 53                     | 0                | 790                    | 6                | None                | 1.0                   | -0.0                |
| 359      | 53                     | 0                | 791                    | 5                | None                | 1.0                   | -0.0                |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value | neg_log10_p_value |
|----------|------------------------|------------------|------------------------|------------------|------------|---------|-------------------|
| 360      | 53                     | 0                | 792                    | 4                | None       | 1.0     | -0.0              |
| 435      | 53                     | 0                | 793                    | 3                | None       | 1.0     | -0.0              |
| 450      | 53                     | 0                | 795                    | 1                | None       | 1.0     | -0.0              |

## domain\_disruption tables

*frame\_domain\_contingency\_table*

|     |     |
|-----|-----|
| 237 | 556 |
| 23  | 33  |

*domain\_disruption\_descriptives*

| Domain_key              | Domain_name                             | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-------------------------|-----------------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_disruption_pf15494 | Scavenger receptor cysteine-rich domain | 849                     | 56                   | 4               | 789              | 0.0036087038179818086                           |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|     |     |
|-----|-----|
| 23  | 30  |
| 237 | 559 |

*domain\_retention\_descriptives*

| Domain_key      | Domain_name | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-----------------|-------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_key_domain | Trypsin     | 849                     | 53                   | 7               | 789              | 0.0046301375875008235                           |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 9    | 57                   | 867               | 0.0657439446366782 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| ABCC4        | 1           |
| ABCG1        | 1           |
| ACP3         | 1           |
| ADHFE1       | 1           |
| AFDN         | 1           |
| AIRE         | 1           |
| ARF4         | 1           |
| ARHGAP26     | 1           |
| ARRB1        | 1           |
| BACH1        | 1           |
| BRAF         | 4           |
| BRCA2        | 1           |
| C2CD2        | 1           |
| CASZ1        | 1           |
| CCN6         | 1           |
| CERS6        | 1           |
| CHAF1B       | 1           |
| CHD3         | 3           |
| CHST12       | 1           |
| CSK          | 1           |
| CYP3A43      | 1           |
| CYP4Z1       | 1           |
| DENND3       | 1           |
| DMD          | 1           |
| DSCAM        | 3           |
| DYM          | 1           |
| DYRK1A       | 1           |
| ELAPOR1      | 1           |
| ENPP6        | 1           |
| ERG          | 764         |
| ETV1         | 14          |
| ETV4         | 3           |
| ETV5         | 4           |
| FIRRM        | 1           |
| FOXP1        | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| GCOM1        | 1           |
| KLK3         | 1           |
| KRTAP10-4    | 1           |
| LSS          | 1           |
| MAD1L1       | 2           |
| MGA          | 1           |
| MRPS6        | 1           |
| MX1          | 2           |
| MYL5         | 1           |
| NLGN1        | 2           |
| NR3C2        | 2           |
| NSUN4        | 1           |
| OSBPL1A      | 1           |
| OSBPL2       | 1           |
| PALS1        | 1           |
| PAXBP1       | 1           |
| PGM5         | 1           |
| PLCD3        | 1           |
| POLDIP3      | 1           |
| PRDM15       | 1           |
| PREX2        | 1           |
| RBPMS2       | 1           |
| RCN1         | 1           |
| RIPK4        | 3           |
| RSL24D1      | 1           |
| SEPTIN11     | 1           |
| SETD4        | 1           |
| SGMS2        | 1           |
| SH3BGR       | 1           |
| SH3GL3       | 1           |
| SIK1         | 1           |
| SIMC1        | 1           |
| SKIL         | 1           |
| SLC60A1      | 1           |
| TSPAN4       | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| TUT7         | 1           |
| U2AF1        | 1           |
| ZNF827       | 1           |

## Per-event results (results.tsv)

| event_id                        | sample_id                 | patient_id | fusion_name        | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                    | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                 | source_site_2_effect_on_frame |
|---------------------------------|---------------------------|------------|--------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------------------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>4522-T01-I<br>M6-1 | P-0014<br>522-T0<br>1-IM6 |            | ABCC4-<br>TMPRSS2  | ABCC4        | out-of-frame | three_prime | 42873289           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |              |                   | ABCC4 (NM_005845) - TMPRSS2 (NM_001135099)<br>Rearrangement : t(13;21)(q32.1;q22.3)(chr13:g.95689572::chr21:g.42873289)   Protein fusion: out of frame (ABCC4-TMPRSS2) | NA                            |
| EVT-P-004<br>3583-T01-I<br>M6-2 | P-0043<br>583-T0<br>1-IM6 |            | ADHFE1-<br>TMPRSS2 | ADHFE1       | in-frame     | three_prime | 42875000           | 1               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |              |                   | ADHFE1 (NM_144650) - TMPRSS2 (NM_001135099)<br>rearrangement: t(8;21)(q13.1;q22.3)(chr8:g.67354656::chr21:g.42875000)   Protein Fusion: in frame {ADHFE1:TMPRSS2}      | NA                            |
| EVT-P-006<br>3444-T01-I<br>M7-3 | P-0063<br>444-T0<br>1-IM7 |            | AIRE-T<br>MPRSS2   | AIRE         | out-of-frame | three_prime | 42861585           | 4               | 117                         | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |              |                   | AIRE (NM_000383) - TMPRSS2 (NM_001135099)<br>rearrangement: c.1279-222: AIRE_c.350-65:TMPRSS2inv   Protein Fusion: out of frame {AIRE:TMPRSS2}                         | NA                            |
| EVT-P-005<br>4154-T01-I<br>M6-5 | P-0054<br>154-T0<br>1-IM6 |            | BACH1-<br>TMPRSS2  | BACH1        | in-frame     | three_prime | 42866728           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |              |                   | BACH1 (NM_001186) - TMPRSS2 (NM_001135099)<br>rearrangement: c.1776+883: BACH1_c.127-223:TMPRSS2inv   Protein Fusion: in frame {BACH1:TMPRSS2}                         | NA                            |
| EVT-P-006<br>9200-T01-I<br>M7-6 | P-0069<br>200-T0<br>1-IM7 |            | BRCA2-<br>TMPRSS2  | BRCA2        | unknown      | three_prime | 42866350           | 3               | 94                          | False                  | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |              |                   | BRCA2 (NM_000059) - TMPRSS2 (NM_001135099)<br>rearrangement: t(13;21)(q13.1;q22.3)(chr13:g.32945390::chr21:g.42866350)   Protein Fusion: mid-exon {BRCA2:TMPRSS2}      | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name        | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                    | lost_domains                                     | disrupted_domains                       | source_annotation_text                                                                                                                                              | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|--------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------------------------------|--------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>1268-T01-I<br>M7-7  | P-0061<br>268-T0<br>1-IM7 |            | C2CD2-<br>TMPRSS2  | C2CD2        | in-frame     | three_prime | 42867076           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | C2CD2 (NM_015500) - TMPRSS2 (NM_001135099) rearrangement: c.280-4023: C2CD2_c.127-571:TMPRSS2del   Protein Fusion: in frame {C2CD2:TMPRSS2}                         | NA                           |
| EVT-P-005<br>8455-T01-I<br>M6-8  | P-0058<br>455-T0<br>1-IM6 |            | CHAF1B-<br>TMPRSS2 | CHAF1B       | in-frame     | three_prime | 42873246           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | CHAF1B (NM_005441) - TMPRSS2 (NM_001135099) rearrangement: c.482-305:CHAF1B_c.56-3130:TMPRSS2inv   Protein Fusion: in frame {CHAF1B:TMPRSS2}                        | NA                           |
| EVT-P-000<br>0449-T02-I<br>M5-10 | P-0000<br>449-T0<br>2-IM5 |            | DSCAM-<br>TMPRSS2  | DSCAM        | out-of-frame | five_prime  | 42851224           | 7               | 228                         | True                   | lost          | 0.0                      | False               |                                                     | Trypsin                                          | Scavenger receptor cysteine-rich domain | TMPRSS2 (NM_001135099) - DSCAM (NM_001389) rearrangement: c.684-15:TMPRSS2_c.509-91895:DSCAMdel   Protein fusion: out of frame (TMPRSS2-DSCAM)                      | NA                           |
| EVT-P-000<br>5459-T01-I<br>M5-11 | P-0005<br>459-T0<br>1-IM5 |            | DSCAM-<br>TMPRSS2  | DSCAM        | out-of-frame | five_prime  | 42867909           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                     | Trypsin; Scavenger receptor cysteine-rich domain |                                         | TMPRSS2 (NM_001135099) - DSCAM (NM_001389) Rearrangement : c.127-1404:TMPRSS2_c.3562+3890:DSCAMdel   Protein fusion: out of frame (TMPRSS2-DSCAM)                   | NA                           |
| EVT-P-006<br>3060-T02-I<br>M7-12 | P-0063<br>060-T0<br>2-IM7 |            | ELAPOR1-TMPRSS2    | ELAPOR1      | in-frame     | three_prime | 42867232           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | KIAA1324 (NM_020775) - TMPRSS2 (NM_001135099) rearrangement: t(1;21)(p13.3;q22.3)(chr1:g.109667031::chr21:g.42867232)   Protein Fusion: in frame {KIAA1324:TMPRSS2} | NA                           |
| EVT-P-006<br>5940-T01-I<br>M7-13 | P-0065<br>940-T0<br>1-IM7 |            | ENPP6-<br>TMPRSS2  | ENPP6        | out-of-frame | three_prime | 42873641           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | ENPP6 (NM_153343) - TMPRSS2 (NM_001135099) rearrangement: t(4;21)(q35.1;q22.3)(chr4:g.185022680::chr21:g.42873641)   Protein Fusion: out of frame {ENPP6:TMPRSS2}   | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                           | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>2132-T02-I<br>M6-14 | P-0012<br>132-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428685<br>21       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10): c.126+<br>1525:TMPRSS2_c.18+2872<br>:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>7312-T01-I<br>M6-15 | P-0017<br>312-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428685<br>23       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon2 with ERG<br>exon 2) : c.126+1523:TMPR<br>SS2_c.18+7805:ERGdel  <br>Protein Fusion: in frame<br>{TMPRSS2:ERG}             | NA                           |
| EVT-P-003<br>6545-T01-I<br>M6-16 | P-0036<br>545-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428705<br>80       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.56-4<br>64:TMPRSS2_c.40-59641:E<br>RGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                | NA                           |
| EVT-P-000<br>6815-T01-I<br>M5-17 | P-0006<br>815-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 428750<br>44       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exon 2) : c.55+48<br>33:TMPRSS2_c.-150+8005:<br>ERGdel   Transcript fusion<br>(TMPRSS2-ERG)            | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>4796-T01-I<br>M7-18 | P-0064<br>796-T0<br>1-IM7 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | three_prime | 42838080           | 14              | 527                         | False                  | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.18+10758:ERG_c.1579:TMPRSS2dup   Protein Fusion: mid-exon {ERG:TMPRSS2}                                     | NA                           |
| EVT-P-000<br>1242-T05-I<br>M6-20 | P-0001<br>242-T0<br>5-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867993           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1-2 with ERG exons 2-10) : c.127-1488:TMPRSS2_c.18+9956:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                           |
| EVT-P-000<br>1697-T02-I<br>M3-21 | P-0001<br>697-T0<br>2-IM3 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872306           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                                                       | NA                           |
| EVT-P-000<br>1698-T01-I<br>M3-22 | P-0001<br>698-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874789           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Deletion (3032220 bp);(TMPRSS2:c.56-4673_ERG:c.19-25025del)   Protein fusion: out of frame (TMPRSS2-ERG)                     | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                        | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------|---------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>8798-T01-I<br>M6-23 | P-0018<br>798-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428722<br>71       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                         | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10): c.56-215<br>5:TMPRSS2_c.19-19621:ER<br>Gdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}      | NA                           |
| EVT-P-000<br>1776-T01-I<br>M3-24 | P-0001<br>776-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | three_prime | 428699<br>80       | 2               | 42                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger<br>receptor cysteine-<br>rich domain |                                                               |                   | NA   Protein fusion: in frame<br>(ERG-TMPRSS2)                                                                                                                                                         | NA                           |
| EVT-P-000<br>1845-T01-I<br>M3-25 | P-0001<br>845-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428671<br>31       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                         | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Deletion<br>(2939631bp): (TMPRSS2:c.<br>127-626_ERG:c.39+20086d<br>el)   Protein fusion: in frame<br>(TMPRSS2-ERG)                                      | NA                           |
| EVT-P-000<br>1876-T02-I<br>M5-26 | P-0001<br>876-T0<br>2-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428706<br>13       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                         | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion (Exon<br>1 of TMPRSS2 fused to<br>exons 5-10 of ERG): c.236+<br>6602:ERG_c.56-497:TMPR<br>SS2del   Protein fusion: out<br>of frame (TMPRSS2-ERG) | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                                     | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>1876-T03-I<br>M5-27 | P-0001<br>876-T0<br>3-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428706<br>13          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion (Exon<br>1 of TMPRSS2 fused to<br>exons 5-10 of ERG): c.56-4<br>97:TMPRSS2_c.236+6602:<br>ERGdel   Protein fusion: out<br>of frame (TMPRSS2-ERG)     | NA                            |
| EVT-P-000<br>1242-T04-I<br>M6-28 | P-0001<br>242-T0<br>4-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428679<br>93          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1-2 with<br>ERG exons 2-10) : c.127-14<br>88:TMPRSS2_c.18+9956:E<br>RGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}               | NA                            |
| EVT-P-000<br>2101-T02-I<br>M6-30 | P-0002<br>101-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428709<br>09          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11):c.56-793:<br>TMPRSS2_c.39+51472:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}            | NA                            |
| EVT-P-000<br>1242-T03-I<br>M5-31 | P-0001<br>242-T0<br>3-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428679<br>92          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exons 1 to 2<br>fused with ERG exons 2 to<br>10) : c.127-1487:TMPRSS2<br>_c.18+9956:ERGdel  <br>Protein fusion: in frame<br>(TMPRSS2-ERG) | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                              | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0002149-T02-I M3-32 | P-0002149-T02-IM3 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42868376           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Deletion : c.126+1670:TMPRSS2_c.18+2482:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)    | NA                           |
| EVT-P-0002206-T01-I M3-33 | P-0002206-T01-IM3 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869351           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: in frame (TMPRSS2-ERG)                                                                                         | NA                           |
| EVT-P-0002296-T01-I M3-35 | P-0002296-T01-IM3 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42877394           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Deletion : c.55+2483:TMPRSS2_c.18+2362:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0002364-T01-I M3-36 | P-0002364-T01-IM3 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872677           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                     | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_doma_in | lost_doma_in                                                                                 | disrupt_ed_doma_in | source_annotation_text                                                                                                                                      | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------|----------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>2372-T02-I<br>M6-37 | P-0002<br>372-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428728<br>15          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.56-2<br>699:TMPRSS2_c.40-60926:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}          | NA                            |
| EVT-P-000<br>2375-T01-I<br>M3-38 | P-0002<br>375-T0<br>1-IM3 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428705<br>85          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | NA   Protein fusion: out of<br>frame (TMPRSS2-ERG)                                                                                                          | NA                            |
| EVT-P-000<br>2375-T02-I<br>M6-39 | P-0002<br>375-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428705<br>86          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918)<br>rearrangement: c.56-470:T<br>MPRSS2_c.19-14125:ERGd<br>el   Protein Fusion: out of<br>frame {TMPRSS2:ERG} | NA                            |
| EVT-P-000<br>2397-T01-I<br>M3-40 | P-0002<br>397-T0<br>1-IM3 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428773<br>64          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Deletion : c.5<br>5+2513:TMPRSS2_c.237-5<br>242:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG)    | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                             | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>9191-T01-I<br>M7-41 | P-0059<br>191-T0<br>1-IM7 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | three_prime | 42861642           | 4               | 117                         | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.18+2242:ERG_c.350-122:TMPRSS2 dup   Protein Fusion: out of frame {ERG:TMPRSS2}           | NA                           |
| EVT-P-000<br>1141-T01-I<br>M3-42 | P-0001<br>141-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42878671           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Deletion (3028438bp): (TMPRSS2:c.55+1206_ERG:c.18+20054 del)   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-000<br>1128-T01-I<br>M3-44 | P-0001<br>128-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42870060           | 2               | 38                          | False                  | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Deletion (3032278 bp): c.112_c.19-20238del   Protein fusion: mid-exon (TMPRSS2-ERG)                       | NA                           |
| EVT-P-000<br>2880-T01-I<br>M3-46 | P-0002<br>880-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867024           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: in frame (TMPRSS2-ERG)                                                                                                        | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role  | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_domains                                                           | lost_do_mains                                                                                | disrupt_ed_domains | source_annotation_text                                                                                                                                       | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|---------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>2894-T01-I<br>M3-47 | P-0002<br>894-T0<br>1-IM3 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv_e_pri_me  | 428683<br>77          | 2                | 42                           | True                     | lost           | 0.0                        | False                |                                                                             | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | NA   Protein fusion: in frame<br>(TMPRSS2-ERG)                                                                                                               | NA                            |
| EVT-P-000<br>2895-T01-I<br>M3-48 | P-0002<br>895-T0<br>1-IM3 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv_e_pri_me  | 428712<br>76          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                                                                             | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | NA   Protein fusion: out of<br>frame (TMPRSS2-ERG)                                                                                                           | NA                            |
| EVT-P-000<br>2901-T01-I<br>M3-49 | P-0002<br>901-T0<br>1-IM3 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv_e_pri_me  | 428690<br>47          | 2                | 42                           | True                     | lost           | 0.0                        | False                |                                                                             | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | NA   Protein fusion: in frame<br>(TMPRSS2-ERG)                                                                                                               | NA                            |
| EVT-P-005<br>2718-T01-I<br>M6-50 | P-0052<br>718-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | thr_ee_pri_me | 428605<br>88          | 5                | 146                          | True                     | retain<br>ed   | 1.0                        | False                | Trypsi<br>n; Sca<br>venger<br>recept<br>or cyst<br>eine-ri<br>ch do<br>main |                                                                                              |                    | ERG (NM_182918) -<br>TMPRSS2<br>(NM_001135099)<br>rearrangement: c.19-20653:<br>ERG_c.437-148:TMPRSS2<br>dup   Protein Fusion: out of<br>frame {ERG:TMPRSS2} | NA                            |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0025332-T01-IM6-51 | P-0025332-T01-IM6 |            | ERG-TMPRSS2 | ERG          | unknown      | five_prime  | 42875781           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11); c.55+4096:TMPRSS2_c.40-55162:ERGdel   Transcript Fusion {TMPRSS2:ERG}            | NA                           |
| EVT-P-0051756-T01-IM6-52 | P-0051756-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | three_prime | 42871205           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.18+346:ERG_c.56-1089:TMPRSS2dup   Protein Fusion: out of frame {ERG:TMPRSS2}                                     | NA                           |
| EVT-P-0002971-T02-IM5-53 | P-0002971-T02-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42871462           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_001136154) Fusion (TMPRSS2 exon 2 fused to ERG exon 4); c.127-1520:TMPRSS2_c.39+27766:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0002990-T01-IM3-54 | P-0002990-T01-IM3 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42869167           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: in frame (TMPRSS2-ERG)                                                                                                                                | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                           | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0002990-T02-I M5-55 | P-0002990-T02-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869167           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 with ERG exons 4-11) : :TMPRSS2_c.40-63033:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                    | NA                           |
| EVT-P-0002990-T03-I M6-56 | P-0002990-T03-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869167           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1 to 2 fused to ERG exons 4 to 10) : c.126+879:TMPRSS2_c.40-63033:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0003023-T01-I M3-59 | P-0003023-T01-IM3 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42874410           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                                                                  | NA                           |
| EVT-P-0003052-T01-I M5-60 | P-0003052-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872061           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                                                                  | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                           | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0003052-T02-IM6-61 | P-0003052-T02-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872062           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exons 1-3 fused with ERG exons 4-11): c.56-1945:TMPRSS2_c.40-5820 4:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-0003139-T01-IM5-62 | P-0003139-T01-IM5 |            | ERG-TM PRSS2 | ERG          | unknown      | five_prime  | 42866368           | 3               | 88                          | False                  | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: mid-exon (TMPRSS2-ERG)                                                                                                                                      | NA                           |
| EVT-P-0003139-T02-IM5-63 | P-0003139-T02-IM5 |            | ERG-TM PRSS2 | ERG          | unknown      | five_prime  | 42866368           | 3               | 88                          | False                  | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-3 with ERG exons 4-10) : c.264:TMPRSS2_c.40-59220:ERGdel   Protein fusion: mid-exon (TMPRSS2-ERG)               | NA                           |
| EVT-P-0003226-T01-IM5-64 | P-0003226-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42874729           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                                                                  | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                    | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0025676-T01-IM6-65 | P-0025676-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42876910           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11);c.55+2967:TMPRSS2_c.39+20578:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-0003419-T02-IM5-66 | P-0003419-T02-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42877994           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | c.40-58284:ERG_c.55+1883:TMPRSS2del   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                          | NA                           |
| EVT-P-0025319-T01-IM6-67 | P-0025319-T01-IM6 |            | ERG-TM PRSS2 | ERG          | unknown      | five_prime  | 42870723           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11);c.56-607:TMPRSS2_c.40-55236:ERGdel   Transcript Fusion {TMPRSS2:ERG}             | NA                           |
| EVT-P-0003511-T01-IM5-68 | P-0003511-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42878695           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                                                           | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0003511-T02-I M5-69 | P-0003511-T02-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42878695           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 4-11): c.39+52408:ERG_c.55+1182:TMPRSS2del   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0003511-T03-I M5-70 | P-0003511-T03-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42878695           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 4-11): c.39+52408:ERG_c.55+1182:TMPRSS2del   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0003535-T01-I M5-71 | P-0003535-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42873785           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_001136154) fusion : c.56-3669:TMPRSS2_c.39+60775:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                 | NA                           |
| EVT-P-0003541-T01-I M5-72 | P-0003541-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42867573           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_001136154) fusion : c.127-1068:TMPRSS2_c.40-44654:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                                    | NA                           |

| event_id                 | sampl_e_id        | pa_tie_nt_id | fusion_name  | part_ner_gene | fra_me_status | tar_get_role | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                    | disrupt_ed_do_mains | source_annotation_text                                                                                                                                             | source_site_2_effect_on_frame |
|--------------------------|-------------------|--------------|--------------|---------------|---------------|--------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|--------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0003592-T01-IM5-73 | P-0003592-T01-IM5 |              | ERG-TM PRSS2 | ERG           | in-frame      | five_prime   | 42866560              | 3                | 43                           | True                     | lost           | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_001243428) fusion: c.127-55:TMPRSS2_c.40-16879:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                                    | NA                            |
| EVT-P-0003592-T02-IM5-74 | P-0003592-T02-IM5 |              | ERG-TM PRSS2 | ERG           | in-frame      | five_prime   | 42866560              | 3                | 43                           | True                     | lost           | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 2 fused to ERG exon 4) : c.127-55:TMPRSS2_c.19-16879:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                            |
| EVT-P-0003597-T01-IM5-75 | P-0003597-T01-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-frame  | five_prime   | 42871994              | 2                | 19                           | True                     | lost           | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-1878:TMPRSS2_c.18+11694:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                  | NA                            |
| EVT-P-0003597-T02-IM5-76 | P-0003597-T02-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-frame  | five_prime   | 42871994              | 2                | 19                           | True                     | lost           | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-1878:TMPRSS2_c.18+11694:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                  | NA                            |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0003602-T01-IM5-77 | P-0003602-T01-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42876381           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | c.55+3496:TMPRSS2_c.39+21260:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                         | NA                           |
| EVT-P-0003619-T01-IM5-78 | P-0003619-T01-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42869579           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon2 with ERG exons 4-11) : c.126+467:TMPRSS2_c.39+43704:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)       | NA                           |
| EVT-P-0003626-T01-IM5-79 | P-0003626-T01-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42867148           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 3 fuses with ERG exons 4-10): c.127-643:ERG_c.40-61697:TMPRSS2del   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0003678-T02-IM6-80 | P-0003678-T02-IM6 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42867219           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | (NM_004449) Fusion (TMPRSS2 exon2 fused with ERG exon4) : c.127-714:TMPRSS2_c.40-61240:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}                                   | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0003689-T01-I M5-81 | P-0003689-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42877684           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | c.55+2193:TMPRSS2_c.18+716:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                               | NA                           |
| EVT-P-0003758-T01-I M5-82 | P-0003758-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42873606           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | c.56-3490:TMPRSS2_c.39+45416:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                                                                             | NA                           |
| EVT-P-0003908-T01-I M3-83 | P-0003908-T01-IM3 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42867649           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | NA   Protein fusion: in frame (TMPRSS2-ERG)                                                                                                                                  | NA                           |
| EVT-P-0003971-T02-I M5-84 | P-0003971-T02-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42875798           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fuses with ERG exons 4-10); c.55+4079:TMPRSS2_c.19-16151:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                    | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0004221-T01-I M5-85 | P-0004221-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42868025           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 2 fused to ERG exon 3): c.127-1520:TM PRSS2_c.237-8364:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                           |
| EVT-P-0004283-T01-I M5-86 | P-0004283-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42875256           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 2-10) : c.55+4621:TM PRSS2_c.18+7946:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)    | NA                           |
| EVT-P-0025150-T01-I M6-87 | P-0025150-T01-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42867731           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 3-10); c.127-1226:TM PRSS2_c.18+10530: ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                           |
| EVT-P-0004285-T01-I M5-88 | P-0004285-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872646           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 2-10) : c.56-2530:TM PRSS2_c.18+10159:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)   | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                           | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0004297-T02-I M5-89 | P-0004297-T02-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869427           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in frame with ERG exons 4-10): c.126+619:TMPRSS2_c.-867:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0004345-T01-I M5-91 | P-0004345-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42870767           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10) : c.56-651:TMPRSS2_c.18+6726:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)      | NA                           |
| EVT-P-0004532-T01-I M5-92 | P-0004532-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869942           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2-ERG Fusion (TMPRSS2 exon 2 fused to ERG exon 4) : c.126+104:TMPRSS2_c.39+63265:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                                           | NA                           |
| EVT-P-0004539-T01-I M5-93 | P-0004539-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42876969           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fuses with ERG exons 2-10): c.55+2908:TMPRSS2_c.40-50360:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)     | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                        | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>5139-T01-I<br>M6-94 | P-0025<br>139-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872490           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.-56-2374:TMPRSS2_c.39+3700:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                             | NA                            |
| EVT-P-000<br>4816-T01-I<br>M5-95 | P-0004<br>816-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873903           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fuses with ERG exons 2-10): c.56-3787:TMPRSS2_c.39+54953:ERGdel.   Protein fusion: out of frame (TMPRSS2-ERG) | NA                            |
| EVT-P-000<br>4826-T01-I<br>M5-96 | P-0004<br>826-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42871052           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 with ERG exons 2-10) : c.56-936:TMPRSS2_c.-150+18179:ERGdel   Transcript fusion (TMPRSS2-ERG)                 | NA                            |
| EVT-P-002<br>5137-T01-I<br>M6-97 | P-0025<br>137-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42869679           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 2 fused with ERG exon 3): c.126+367:TMPRSS2_c.237-934:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}       | NA                            |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                         | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0005064-T01-I M5-99  | P-0005064-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42870659           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2(NM_001135099) -ERG (NM_182918) Fusion (TMPRSS2 exon1 ERG exon2 ) : c.56-543:TMPRSS2_c.18+11495:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                     | NA                           |
| EVT-P-0005098-T02-I M5-101 | P-0005098-T02-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42868037           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exons 1-2 fused with ERG exons 2-10 ) : c.127-1532:TMPRSS2_c.18+25709:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0005422-T01-I M5-102 | P-0005422-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42875709           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2(NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon1 ERG exon4 ) : c.55+4168:TMPRSS2_c.40-60922:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                   | NA                           |
| EVT-P-0025823-T01-I M6-104 | P-0025823-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42870362           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused to ERG exons 2-10): c.56-246:TMPRSS2_c.18+2365:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}    | NA                           |

| event_id                   | sampl_e_id        | pa_tie_nt_id | fusion_name  | part_ner_gene | fra_me_status | tar_get_role | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_doma_in | lost_doma_in                                             | disrupt_ed_doma_in | source_annotation_text                                                                                                                                                  | source_site_2_effect_on_frame |
|----------------------------|-------------------|--------------|--------------|---------------|---------------|--------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------|----------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0005454-T01-I M5-105 | P-0005454-T01-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-frame  | fiv_e_prime  | 42872059              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                    | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion(TMPRSS2 exon 1 fused to ERG exon2 ) : c.56-1943:TMPRSS2_c.18+13593:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                            |
| EVT-P-0005454-T02-I M5-106 | P-0005454-T02-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-frame  | fiv_e_prime  | 42872059              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                    | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-1943:TMPRSS2_c.18+13593:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                       | NA                            |
| EVT-P-000888-T01-I M3-107  | P-0000888-T01-IM3 |              | ERG-TM PRSS2 | ERG           | in-frame      | fiv_e_prime  | 42868672              | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                   | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                    | TMPRSS2 (NM_001135099) - ERG (NM_182918) Deletion: c.126+1374_c.19-17794del   Protein fusion: in frame (TMPRSS2-ERG)                                                    | NA                            |
| EVT-P-0025851-T01-I M6-108 | P-0025851-T01-IM6 |              | ERG-TM PRSS2 | ERG           | in-frame      | fiv_e_prime  | 42869296              | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                   | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                    | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 2 fused to ERG exons 4-11):c.c.126+750:TMPRSS2_c.40-60753:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_i_d | fusion_name     | part_ner_ge_ne | fra_me_st_atu_s  | tar_get_r_ole | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                        | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|---------------|-----------------|----------------|------------------|---------------|-----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>5463-T01-I<br>M5-109 | P-0005<br>463-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | in-fr<br>ame     | fiv_e_pri_me  | 428680<br>71          | 3                 | 43                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon2 fused to<br>ERG exon 4) :<br>c.127-1566:TMPRSS2<br>_c.40-59106:ERGdel  <br>Protein fusion: in frame<br>(TMPRSS2-ERG)  | NA                            |
| EVT-P-000<br>5546-T01-I<br>M5-110 | P-0005<br>546-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | in-fr<br>ame     | fiv_e_pri_me  | 428685<br>55          | 2                 | 42                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 and<br>ERG exons 2-10) : c.126+14<br>91:TMPRSS2_c.19-14111:E<br>RGdel   Protein fusion: in<br>frame (TMPRSS2-ERG) | NA                            |
| EVT-P-000<br>5557-T01-I<br>M5-111 | P-0005<br>557-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | out-of-fr<br>ame | fiv_e_pri_me  | 428739<br>26          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon1 with ERG<br>exons 2-10) : c.56-3810:TM<br>PRSS2_c.19-16939:ERGdel<br>  Protein fusion: out of frame<br>(TMPRSS2-ERG)  | NA                            |
| EVT-P-000<br>5571-T01-I<br>M5-112 | P-0005<br>571-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | out-of-fr<br>ame | fiv_e_pri_me  | 428775<br>27          | 1                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon1 fused<br>with ERG exon4) : c.55+235<br>0:TMPRSS2_c.40-58808:E<br>RG   Protein fusion: out of<br>frame (TMPRSS2-ERG)   | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_d_o_mains | lost_d_o_mains                                                                               | disrupt_ed_d_o_mains | source_annotation_text                                                                                                                                                                         | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------|----------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>0479-T01-I<br>M3-113 | P-0000<br>479-T0<br>1-IM3 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428726<br>80          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | NA   Protein fusion: out of<br>frame (TMPRSS2-ERG)                                                                                                                                             | NA                            |
| EVT-P-000<br>5636-T01-I<br>M5-114 | P-0005<br>636-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428751<br>16          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon1 fused<br>with ERG exon2) : c.55+476<br>1:TMPRSS2_c.19-17111:E<br>RGdel   Protein fusion: out<br>of frame (TMPRSS2-ERG) | NA                            |
| EVT-P-000<br>5636-T02-I<br>M6-115 | P-0005<br>636-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428751<br>17          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.55+4<br>760:TMPRSS2_c.19-17111:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                             | NA                            |
| EVT-P-000<br>5698-T01-I<br>M5-117 | P-0005<br>698-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428696<br>50          | 2                | 42                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 with<br>ERG exons 4-10) : c.126+39<br>6:TMPRSS2_c.39+21608:E<br>RGdel   Protein fusion: in<br>frame (TMPRSS2-ERG)  | NA                            |

| event_id                   | sampl_e_id        | pa_tie_nt_id | fusion_name  | part_ner_gene | fra_me_status | tar_get_role | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_d_o_mains | lost_d_o_mains                                           | disrupt_ed_d_o_mains | source_annotation_text                                                                                                                                                            | source_site_2_effect_on_frame |
|----------------------------|-------------------|--------------|--------------|---------------|---------------|--------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------|----------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0005756-T01-I M5-118 | P-0005756-T01-IM5 |              | ERG-TM PRSS2 | ERG           | in-fr_ame     | fiv_e_pri_me | 42867719              | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                     | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                      | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 with ERG exons 2-10) : c.127-1214:TMPRSS2_c.18+26157: ERGdel   Protein fusion: in frame (TMPRSS2-ERG)          | NA                            |
| EVT-P-0005864-T01-I M5-120 | P-0005864-T01-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-fr_ame | fiv_e_pri_me | 42874935              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                      | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 4-11): c.56-4819:TMP RSS2_c.19-12436:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)            | NA                            |
| EVT-P-0025097-T01-I M6-121 | P-0025097-T01-IM6 |              | ERG-TM PRSS2 | ERG           | out-of-fr_ame | fiv_e_pri_me | 42875596              | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                      | TMPRSS2 (NM_001135099) - ERG (NM_004449) rearrangement (TMPRSS2 exon 1 fused to ERG exons 4-11): c.55+4281:TMPRSS2_c.40-57110:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                            |
| EVT-P-0005953-T01-I M5-122 | P-0005953-T01-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-fr_ame | fiv_e_pri_me | 42876716              | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                      | TMPRSS2 (NM_001135099) -ERG (NM_182918) Fusion (TMPRSS2 exon 1 fused to ERG exons 3 to 10) : c.55+3161:TMPRSS2_c.237-9105: ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)    | NA                            |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                                  | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0005953-T02-I M5-123 | P-0005953-T02-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42876716           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) reciprocal fusion (TMPRSS2 exon1 fused with ERG exons 5-11) : c.55+3161:TMPRSS2_c.258-9105:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0005953-T03-I M6-124 | P-0005953-T03-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42876716           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 3-10): c.55+3161:TMPRSS2_c.237-9105:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)              | NA                           |
| EVT-P-0006108-T01-I M5-126 | P-0006108-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869746           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 with ERG exons 2-10) : c.126+300:TMPRSS2_c.18+8285:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                   | NA                           |
| EVT-P-0036791-T01-I M6-127 | P-0036791-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42873619           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-3503:TMPRSS2_c.18+1131:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                        | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                            | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0006108-T02-I M5-128 | P-0006108-T02-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869746           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) fusion (TMPRSS2 exons 1 and 2 fused with ERG exons 3 to 10) : c.126+300:TMPRSS2_c.18+8285:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0036756-T01-I M6-129 | P-0036756-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42877290           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+2587:TMPRSS2_c.19-4454:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                  | NA                           |
| EVT-P-0036699-T01-I M6-130 | P-0036699-T01-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42868238           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.127-1733:TMPRSS2_c.40-58568:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}                                                    | NA                           |
| EVT-P-0025044-T01-I M6-131 | P-0025044-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872368           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon1 fused with ERG exon 2) : c.56-2252:TMPRSS2_c.18+17253:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}          | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status    | tar_get_role | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_d_o_mains | lost_d_o_mains                                                                               | disrupt_ed_d_o_mains | source_annotation_text                                                                                                                                                                             | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|------------------|--------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------|----------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>6602-T02-I<br>M5-132 | P-0006<br>602-T0<br>2-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame     | fiv_e_pri_me | 428670<br>72          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 with<br>ERG exons 4-11) : c.127-56<br>7:TMPRSS2_c.40-17587:E<br>RGdel   Protein fusion: in<br>frame (TMPRSS2:ERG)      | NA                            |
| EVT-P-000<br>6602-T03-I<br>M6-133 | P-0006<br>602-T0<br>3-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame     | fiv_e_pri_me | 428670<br>73          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 3-10): c.12<br>7-568:TMPRSS2_c.19-1758<br>7:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG} | NA                            |
| EVT-P-002<br>5030-T01-I<br>M6-134 | P-0025<br>030-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri_me | 428776<br>33          | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 with<br>ERG exons 2-10): c.55+224<br>4:TMPRSS2_c.19-2444:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}       | NA                            |
| EVT-P-000<br>6619-T01-I<br>M5-136 | P-0006<br>619-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame     | fiv_e_pri_me | 428675<br>71          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 with<br>ERG exons 4-11) : c.127-10<br>66:TMPRSS2_c.40-44616:E<br>RGdel   Protein fusion: in<br>frame (TMPRSS2:ERG)     | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                                    | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>3575-T01-I<br>M6-138 | P-0033<br>575-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428703<br>54          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon2 fused<br>with ERG exon2) : c.56-238:<br>TMPRSS2_c.18+8414:ERG<br>del   Protein Fusion: out of<br>frame {TMPRSS2:ERG}              | NA                            |
| EVT-P-001<br>2741-T01-I<br>M5-139 | P-0012<br>741-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428782<br>72          | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon1 with ERG<br>exons 4-11): c.55+1605:TM<br>PRSS2_c.40-54554:ERGdel<br>  Protein fusion: out of frame<br>(TMPRSS2-ERG)               | NA                            |
| EVT-P-000<br>6837-T01-I<br>M5-140 | P-0006<br>837-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428761<br>81          | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4 to 10) : c.55+3<br>696:TMPRSS2:c.40-56077:<br>ERG_del   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                            |
| EVT-P-003<br>6438-T01-I<br>M6-141 | P-0036<br>438-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428676<br>49          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.127-<br>1144:TMPRSS2_c.18+2770<br>:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}                                                            | NA                            |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_nic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|-------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0006837-T02-I M5-142 | P-0006837-T02-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42876181           | 1               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 1-10); c.55+3696:TMPRSS2_c.-3317:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)         | NA                           |
| EVT-P-0036351-T01-I M6-143 | P-0036351-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872264           | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG(NM_004449) Fusion (TMPRSS2 exon2 with ERG exon4) : c.56-2148:TMPRSS2_c.39+62508:ERG del   Protein Fusion: out of frame {TMPRSS2:ERG}               | NA                           |
| EVT-P-0006840-T01-I M5-144 | P-0006840-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872258           | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon 1 fused to ERG exons 4 to 10) : c.55+3696:TMPRSS2:c.39+61789:ERG_del   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0001319-T03-I M5-145 | P-0001319-T03-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42879131           | 1               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exons 4-11) : c.55+746:TMPRSS2_c.40-60506:ERG_del   Protein fusion: out of frame (TMPRSS2-ERG)   | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                                 | source_site_2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0001319-T05-I M5-146 | P-0001319-T05-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42879131           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exons 4-11): c.55+746:TMPRSS2_c.40-60506:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)            | NA                            |
| EVT-P-0006905-T01-I M5-148 | P-0006905-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42871259           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) reciprocal fusion (TMPRSS exon1 fused with ERG exons 4-11) : c.56-1143:TMPRSS2_c.40-62829:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                            |
| EVT-P-0006346-T01-I M6-149 | P-0006346-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42870600           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon1 fused with ERG exon4): c.56-484:TMPRSS2_c.39+15996:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                  | NA                            |
| EVT-P-0007025-T01-I M5-150 | P-0007025-T01-IM5 |            | ERG-TM PRSS2 | ERG          | unknown      | five_prime  | 42873714           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exons 2-11) : c.56-3598:TMPRSS2_c.-150+4207:ERGdel   Transcript fusion (TMPRSS2-ERG)                    | NA                            |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                        | source_site_2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0007043-T01-I M5-151 | P-0007043-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869028           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 2 fused with ERG exons 2 to 10) : c. 126+1018:TMPRSS2_c.18+3315:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                            |
| EVT-P-0007043-T02-I M6-152 | P-0007043-T02-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869029           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused to ERG exons 2-10) : c.126+1017:TMPRSS2_c.18+3315:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}    | NA                            |
| EVT-P-0036294-T01-I M6-153 | P-0036294-T01-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42867521           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.127-1016:TMPRSS2_c.18+4202:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}                                                 | NA                            |
| EVT-P-0036176-T01-I M6-154 | P-0036176-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42873939           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-3823:TMPRSS2_c.18+14211:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                             | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_nic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                             | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|-------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0025851-T02-IM6-155 | P-0025851-T02-IM6 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42869297           | 2               | 42                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.126+749:TMPRSS2_c.40-60753:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}          | NA                           |
| EVT-P-0007307-T01-IM5-156 | P-0007307-T01-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42869887           | 2               | 42                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1 to 2 fused with ERG exons 2 to 10) : c.126+159:TMPRSS2_c.18+25874:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0007307-T02-IM5-157 | P-0007307-T02-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42869887           | 2               | 42                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons1-2 with ERG exon2-10) : c.126+159:TMPRSS2_c.18+25874:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                | NA                           |
| EVT-P-0007307-T03-IM6-158 | P-0007307-T03-IM6 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42869901           | 2               | 42                          | True                    | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 2 fused to ERG exon2): c.126+145:TMPRSS2_c.18+25873:ERG   Protein Fusion: in frame {TMPRSS2:ERG}                     | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                                 | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0007343-T01-I M5-159 | P-0007343-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869383           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) fusion (TMPRSS2 exons 1 and 2 fused with ERG exons 4 to 11) : c.126+663:TMPRSS2_c.39+64787:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)     | NA                           |
| EVT-P-0007343-T02-I M6-160 | P-0007343-T02-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869384           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 with ERG exons 4-10); c.126+662TMPRSS2_c.39+64787ER Gdel.   Protein Fusion: in frame {TMPRSS2:ERG}                  | NA                           |
| EVT-P-0007359-T01-I M5-162 | P-0007359-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42873736           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) fusion (TMPRSS2 exons 1 and 2 fused with ERG exons 1 to 10) : c.56-3620:TMPRSS2_c.40-55852:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0007366-T01-I M5-163 | P-0007366-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872445           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) fusion (TMPRSS2 exons 1 and 2 fused with ERG exons 2 to 10) : c.56-2329:TMPRSS2_c.40-19374:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0007416-T01-IM5-164 | P-0007416-T01-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42875820           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) fusion (TMPRSS2 exons 1 and fused with ERG exons 2 to 10) : c.55+4057:TMPRSS2_c.-11:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0007416-T02-IM5-165 | P-0007416-T02-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42875820           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 1-10): c.55+4057:TMPRSS2_c.-11:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                | NA                           |
| EVT-P-0007538-T01-IM5-166 | P-0007538-T01-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42876973           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 1 with ERG exons 2 to 10) : c.55+2904:TMPRSS2_c.19-10639:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)      | NA                           |
| EVT-P-0036120-T01-IM6-167 | P-0036120-T01-IM6 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42868693           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused with ERG exons 2-10): c.126+1353:TMPRSS2_c.19-13252:ERGdel   Protein Fusion: in frame (TMPRSS2:ERG)    | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                            | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0007576-T01-I M5-168 | P-0007576-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869584           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused with ERG exons 2 to 10) : c. 126+462:TMPRSS2_c.19-14230:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)  | NA                           |
| EVT-P-0007749-T01-I M5-169 | P-0007749-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42875524           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1 fused with ERG exons 4 to 10): c. 55+4353:TMPRSS2_c.40-54655:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0007770-T01-I M5-170 | P-0007770-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869589           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1-2 fused with ERG exons 4-11) : c.126+457:TMPRSS2_c.39+35604:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)       | NA                           |
| EVT-P-0008077-T01-I M5-172 | P-0008077-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42868270           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4 to 11): c. 127-1765:TMPRSS2_c.39+64394:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)      | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                             | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008139-T01-IM5-173 | P-0008139-T01-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42867120           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused with ERG exons 3-11) : c.127-615:TMPRSS2_c.40-33298:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)       | NA                           |
| EVT-P-0008139-T02-IM5-174 | P-0008139-T02-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42867120           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1 to 2 fused with ERG exons 3 to 11): c.127-615:TMPRSS2_c.40-33298:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)  | NA                           |
| EVT-P-0008139-T03-IM5-175 | P-0008139-T03-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42867120           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in-frame to ERG exons 2-10): c.127-615:TMPRSS2_c.18+19445:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0008147-T01-IM5-176 | P-0008147-T01-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42874297           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) c.56-4181:TMPRSS2_c.40-58022:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)         | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                        | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008206-T01-I M5-177 | P-0008206-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42870501           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4 to 11): c.56-385:TMPRSS2_c.39+41500:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0008232-T01-I M5-178 | P-0008232-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42868959           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused with ERG exons 3-11) : c.126+1087:TMPRSS2_c.40-59158:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0036007-T01-I M6-179 | P-0036007-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42874243           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 1 with ERG exon 2) : c.56-4127:TMPRSS2_c.18+10822:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}           | NA                           |
| EVT-P-0008304-T01-I M5-180 | P-0008304-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42871261           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-1145:TMPRSS2_c.40-48890:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>5854-T01-I<br>M6-181 | P-0035<br>854-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428695<br>13       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.126+<br>533:TMPRSS2_c.39+11353<br>:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}                                                          | NA                           |
| EVT-P-000<br>8383-T01-I<br>M5-182 | P-0008<br>383-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428746<br>65       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11) : c.56<br>-4549:TMPRSS2_c.40-5424<br>5:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                           |
| EVT-P-000<br>8441-T01-I<br>M5-183 | P-0008<br>441-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428681<br>17       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11) : c.12<br>7-1612:TMPRSS2_c.39+60<br>626:ERGdel   Protein fusion:<br>in frame (TMPRSS2-ERG)    | NA                           |
| EVT-P-003<br>5818-T01-I<br>M6-184 | P-0035<br>818-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428677<br>27       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon2 fused<br>with ERG exon2) : c.127-12<br>22:TMPRSS2_c.19-15806:ER<br>Gdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}             | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008452-T01-I M5-185 | P-0008452-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872698           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4 to 11): c.56-2582:TMPRSS2_c.39+45360:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                           |
| EVT-P-0008525-T01-M5-186   | P-0008525-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869423           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exons 1-2 fused with ERG exons 2-11) : c.126+623:TMPRSS2_c.18+8025:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)     | NA                           |
| EVT-P-0008525-T02-I M6-187 | P-0008525-T02-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869424           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 with ERG exons 2-10): c.126+622:TMPRSS2_c.40-44718:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}           | NA                           |
| EVT-P-0008542-T01-I M5-188 | P-0008542-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42875567           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exons 1 fused with ERG exons 2 -10 ) : c.55+4310:TMPRSS2_c.19-7094:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                              | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008591-T01-IM5-189 | P-0008591-T01-IM5 |            | ERG-TMPRSS2 | ERG          | in-frame     | five_prime  | 42868017           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) fusion (TMPRSS2 exons 1 and 2 fused with ERG exons 4 to 11) : c.127-1512:TMPRSS2_c.39+64467:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0008592-T01-IM5-190 | P-0008592-T01-IM5 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42877519           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG(NM_004449) Fusion (TMPRSS2 exon 1 fused with ERG exons 4-11) : c.55+2358:TMPRSS2_c.39+59122:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)        | NA                           |
| EVT-P-0035561-T01-IM6-191 | P-0035561-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42871825           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon2 fused with ERG exon2) : c.56-1709:TMPRSS2_c.18+5540:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}              | NA                           |
| EVT-P-0035506-T01-IM6-192 | P-0035506-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | three_prime | 42878701           | 1               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.19-16831:ERG_c.55+1176:TMPRSS2 dup   Protein Fusion: out of frame {ERG:TMPRSS2}                                           | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_i_d | fusion_name     | part_ner_ge_ne | fra_me_st_atu_s      | tar_get_r_ole          | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|---------------|-----------------|----------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>5445-T01-I<br>M6-193 | P-0035<br>445-T0<br>1-IM6 |               | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428744<br>45          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-4<br>329:TMPRSS2_c.18+15866<br>:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}                                             | NA                            |
| EVT-P-000<br>8812-T01-I<br>M5-194 | P-0008<br>812-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428665<br>28          | 3                 | 43                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon1-2 with<br>ERG exons 2-10): c.127-23:<br>TMPRSS2_c.19-9142:ERG<br>del   Protein fusion: in frame<br>(TMPRSS2-ERG)          | NA                            |
| EVT-P-000<br>8869-T01-I<br>M5-195 | P-0008<br>869-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428723<br>50          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10) : c.56-223<br>4:TMPRSS2_c.19-7662:ER<br>Gdel   Protein fusion: out of<br>frame (TMPRSS2-ERG) | NA                            |
| EVT-P-000<br>8879-T01-I<br>M5-196 | P-0008<br>879-T0<br>1-IM5 |               | ERG-TM<br>PRSS2 | ER<br>G        | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428692<br>42          | 2                 | 42                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 2-10): c.12<br>6+804:TMPRSS2_c.18+212<br>:ERGdel   Protein fusion: in<br>frame (TMPRSS2-ERG)  | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                                       | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>5284-T01-I<br>M6-197 | P-0035<br>284-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869055           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.126+991:TMPRSS2_c.40-63565:<br>ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}                                                                   | NA                           |
| EVT-P-002<br>4903-T01-I<br>M6-199 | P-0024<br>903-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873958           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion : c.56-3842:TMPRSS2_c.40-4079<br>0:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}                                                           | NA                           |
| EVT-P-000<br>8960-T01-I<br>M5-200 | P-0008<br>960-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867649           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1 and 2<br>fused with ERG exons 4 to<br>11): c.127-1144:TMPRSS2_<br>c.39+62016:ERGdel  <br>Protein fusion: in frame<br>(TMPRSS2:ERG) | NA                           |
| EVT-P-000<br>8960-T02-I<br>M6-201 | P-0008<br>960-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867650           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1 and 2<br>fused with ERG exons 4 to<br>11): c.127-1145:TMPRSS2_<br>c.39+62016:ERGdel  <br>Protein Fusion: in frame<br>{TMPRSS2:ERG} | NA                           |

| event_id                   | sampl_e_id        | pa_tie_nt_id | fusion_name  | part_ner_gene | fra_me_status | tar_get_role | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                            | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                        | source_site_2_effect_on_frame |
|----------------------------|-------------------|--------------|--------------|---------------|---------------|--------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0009002-T01-I M5-202 | P-0009002-T01-IM5 |              | ERG-TM PRSS2 | ERG           | unkn_ow_n     | fiv_e_pri_me | 42870306              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp_sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                     | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons1-11) : c.56-190:TMPRSS2_c.-150+102 47:ERGdel   Transcript fusion (TMPRSS2-ERG)            | NA                            |
| EVT-P-0009006-T01-I M5-203 | P-0009006-T01-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-fr_ame | fiv_e_pri_me | 42874595              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp_sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                     | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-4479:TMPRSS2_c.40-5032 9:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                            |
| EVT-P-0009006-T02-I M6-204 | P-0009006-T02-IM6 |              | ERG-TM PRSS2 | ERG           | out-of-fr_ame | fiv_e_pri_me | 42874596              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp_sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                     | TMPRSS2 (NM_001135099) rearrangement: c.56-4480:TMPRSS2_chr21:g.3986787 3del   Protein Fusion: out of frame {TMPRSS2:ERG}                                                     | NA                            |
| EVT-P-0009056-T01-I M5-205 | P-0009056-T01-IM5 |              | ERG-TM PRSS2 | ERG           | out-of-fr_ame | fiv_e_pri_me | 42870979              | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp_sin; Sca ven ger r ece ptor cyst eine -rich dom ain |                     | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-863:TMPRSS2_c.40-40299 :ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                            |

| event_id                   | sampl_e_id        | pa_tie_nt_id | fusion_name  | part_ner_ge_ne | fra_me_st_atu_s | tar_get_r_ole | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                    | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                       | source_site_2_effect_on_frame |
|----------------------------|-------------------|--------------|--------------|----------------|-----------------|---------------|-----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|--------------------|--------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0009056-T02-I M6-206 | P-0009056-T02-IM6 |              | ERG-TM PRSS2 | ERG            | out-of-frame    | five_prime    | 42870980              | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-864:TMPRSS2_c.18+12444:ERGdel   Protein Fusion: out of frame (TMPRSS2:ERG)  | NA                            |
| EVT-P-0009073-T01-I M5-207 | P-0009073-T01-IM5 |              | ERG-TM PRSS2 | ERG            | out-of-frame    | five_prime    | 42873613              | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-3497:TMPRSS2_c.40-8505:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                            |
| EVT-P-0009077-T01-I M5-208 | P-0009077-T01-IM5 |              | ERG-TM PRSS2 | ERG            | out-of-frame    | five_prime    | 42870787              | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-671:TMPRSS2_c.40-59954:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                            |
| EVT-P-0009082-T01-I M5-209 | P-0009082-T01-IM5 |              | ERG-TM PRSS2 | ERG            | out-of-frame    | five_prime    | 42872956              | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Trypsin; Scavenger receptor cysteine-rich domain |                     | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-2840:TMPRSS2_c.40-50568:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                         | disrupted_domains | source_annotation_text                                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>5221-T01-I<br>M6-210 | P-0035<br>221-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874805           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.56-4689:TMPRSS2_c.39+50100:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                     | NA                           |
| EVT-P-000<br>9361-T01-I<br>M5-211 | P-0009<br>361-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867892           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused in frame with ERG exons 4-11): c.127-1387:TMPRSS2_c.40-63140:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-000<br>9374-T02-I<br>M5-212 | P-0009<br>374-T0<br>2-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867657           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons1-2 with ERG exon2-10) : c.127-1152:TMPRSS2_c.18+82:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                     | NA                           |
| EVT-P-003<br>5065-T01-I<br>M6-213 | P-0035<br>065-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875047           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.55+4830:TMPRSS2_c.39+38144:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                     | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>6153-T01-I<br>M6-215 | P-0026<br>153-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868435           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon2 fused to<br>ERG exon4) : c.126+1611:T<br>MPRSS2_c.39+54119:ERG<br>del   Protein Fusion: in frame<br>{TMPRSS2:ERG}             | NA                           |
| EVT-P-000<br>9687-T01-I<br>M5-216 | P-0009<br>687-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876697           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 2-10); c.55<br>+3180:TMPRSS2_c.18+601<br>6:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                           |
| EVT-P-000<br>9687-T02-I<br>M6-217 | P-0009<br>687-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876698           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 2-10); c.55<br>+3180:TMPRSS2_c.18+601<br>6:ERGdel   Protein Fusion:<br>out of frame<br>(TMPRSS2:ERG) | NA                           |
| EVT-P-000<br>9728-T01-I<br>M5-218 | P-0009<br>728-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867606           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11) : c.12<br>7-1101:TMPRSS2_c.40-529<br>49:ERGdel   Protein fusion:<br>in frame (TMPRSS2-ERG)  | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status    | tar_get_role | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                           | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|------------------|--------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>9786-T01-I<br>M5-219 | P-0009<br>786-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame     | fiv_e_pri_me | 428695<br>53          | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2(NM_001135099)<br>-ERG (NM_004449)Fusion<br>(TMPRSS2 exon1 fused with<br>ERG exons 4-10) : c.126+49<br>3:TMPRSS2_c.40-60552:E<br>RGdel   Protein fusion: in<br>frame (TMPRSS2-ERG)        | NA                            |
| EVT-P-000<br>9786-T02-I<br>M5-220 | P-0009<br>786-T0<br>2-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame     | fiv_e_pri_me | 428695<br>53          | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11) : c.12<br>6+493:TMPRSS2_c.40-605<br>52:ERG   Protein fusion: in<br>frame (TMPRSS2-ERG) | NA                            |
| EVT-P-000<br>9787-T01-I<br>M5-221 | P-0009<br>787-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri_me | 428756<br>09          | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2(NM_001135099)<br>-ERG ()Fusion (TMPRS22<br>exon1 fused with ERG<br>exons 4-10) : c.55+4268:TM<br>PRSS2_c.40-58604:ERGdel<br>  Protein fusion: out of frame<br>(TMPRSS2-ERG)              | NA                            |
| EVT-P-000<br>9787-T02-I<br>M5-222 | P-0009<br>787-T0<br>2-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri_me | 428756<br>09          | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+4<br>268:TMPRSS2_c.40-58604:<br>ERGdel   Protein fusion: out<br>of frame (TMPRSS2-ERG)                                               | NA                            |

| event_id                      | sample_id         | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                              | source_site2_effect_on_frame |
|-------------------------------|-------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0009819-T01-I<br>M5-223 | P-0009819-T01-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867985           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused with ERG exons 4-11) : c.127-1480:TMPRSS2_c.40-52376:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)       | NA                           |
| EVT-P-0034865-T01-I<br>M6-224 | P-0034865-T01-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874660           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.56-4544:TMPRSS2_c.40-63678:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                   | NA                           |
| EVT-P-0009892-T01-I<br>M5-225 | P-0009892-T01-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867056           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in frame with ERG exons 2-10): c.127-551:TMPRSS2_c.18+3860:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0010015-T01-I<br>M5-226 | P-0010015-T01-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871490           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 fused with ERG exons 2-9): c.56-1374:TMPRSS2_c.18+8903:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)           | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>0071-T01-I<br>M5-227 | P-0010<br>071-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869039           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1 to 2 fused to ERG exons 3 to 10) : c.126+1007:TMPRSS2_c.19-18011:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)    | NA                            |
| EVT-P-001<br>0659-T01-I<br>M5-228 | P-0010<br>659-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867516           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in frame with ERG exons 2-10): c.127-1011:TMPRSS2_c.18+2079:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                            |
| EVT-P-001<br>0794-T01-I<br>M5-230 | P-0010<br>794-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42870148           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 2-10) : c.56-32:TMPRSS2_c.18+3305:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                  | NA                            |
| EVT-P-001<br>0796-T01-I<br>M5-231 | P-0010<br>796-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871526           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 with ERG exons 2-10) : c.56-1410:TMPRSS2_c.19-20820:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)               | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>0796-T02-I<br>M6-232 | P-0010<br>796-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428715<br>27       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon1 with ERG<br>exons 2-10) : c.56-1410:TM<br>PRSS2_c.19-20820:ERGdel<br>  Protein Fusion: out of<br>frame {TMPRSS2:ERG}           | NA                           |
| EVT-P-001<br>0953-T01-I<br>M5-233 | P-0010<br>953-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428722<br>63       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11): c.56-<br>2147:TMPRSS2_c.40-3776<br>1:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                           |
| EVT-P-001<br>1156-T01-I<br>M5-234 | P-0011<br>156-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428755<br>03       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 2-10): c.55<br>+4374:TMPRSS2_c.18+159<br>05:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                           |
| EVT-P-003<br>4815-T01-I<br>M6-235 | P-0034<br>815-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428731<br>56       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-3<br>040:TMPRSS2_c.18+3720:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                      | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>4674-T01-I<br>M6-236 | P-0034<br>674-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875738           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+4139:TMPRSS2_c.19-24509:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                     | NA                           |
| EVT-P-001<br>1364-T01-I<br>M5-237 | P-0011<br>364-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42877201           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 with ERG exons 4-11) : c.55+2676:TMPRSS2_c.40-58736:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                | NA                           |
| EVT-P-003<br>4620-T01-I<br>M6-238 | P-0034<br>620-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875887           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+3990:TMPRSS2_c.18+9776:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                      | NA                           |
| EVT-P-001<br>1377-T01-I<br>M5-239 | P-0011<br>377-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867668           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in frame with ERG exons 4-10): c.127-1163:TMPRSS2_c.19-14268:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>1429-T01-I<br>M5-240 | P-0011<br>429-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876141           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.55+3736:TMPRSS2_c.40-43016:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-003<br>4480-T01-I<br>M6-241 | P-0034<br>480-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871949           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11) : c.56-1833:TMPRSS2_c.39+53541:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-001<br>1472-T01-I<br>M5-242 | P-0011<br>472-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872511           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (exon 1 of TMPRSS2 with exons 2-10 of ERG): c.56-2395:TMPRSS2_c.18+9580:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                           |
| EVT-P-003<br>4431-T01-I<br>M6-243 | P-0034<br>431-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872774           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-2658:TMPRSS2_c.18+15598:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}   | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                               | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>4384-T01-I<br>M6-244 | P-0034<br>384-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428726<br>22          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-2<br>506:TMPRSS2_c.18+6601:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                    | NA                            |
| EVT-P-001<br>1534-T01-I<br>M5-246 | P-0011<br>534-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428779<br>36          | 1                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exon 4-11): c.55+1941<br>:TMPRSS2_c.39+64333:ER<br>Gdel   Protein fusion: out of<br>frame (TMPRSS2-ERG)     | NA                            |
| EVT-P-001<br>1577-T01-I<br>M5-247 | P-0011<br>577-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428670<br>16          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11) : c.12<br>7-511:TMPRSS2_c.40-4714<br>6:ERGdel   Protein fusion: in<br>frame (TMPRSS2-ERG)  | NA                            |
| EVT-P-001<br>2029-T01-I<br>M5-248 | P-0012<br>029-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428681<br>58          | 3                 | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11) : c.12<br>7-1653:TMPRSS2_c.19-137<br>47:ERGdel   Protein fusion:<br>in frame (TMPRSS2-ERG) | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                              | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                              | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------|---------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>4296-T01-I<br>M6-249 | P-0034<br>296-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428714<br>19       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                               | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.56-1<br>303:TMPRSS2_c.40-58938:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                  | NA                           |
| EVT-P-001<br>2046-T01-I<br>M5-250 | P-0012<br>046-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | three_prime | 428722<br>27       | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                                                               |                   | ERG (NM_004449) -<br>TMPRSS2<br>(NM_001135099)<br>rearrangement : c.40-45795:<br>ERG_c.56-2111:TMPRSS2<br>dup   Protein fusion: out of<br>frame (ERG-TMPRSS2)                                       | NA                           |
| EVT-P-001<br>2048-T01-I<br>M5-251 | P-0012<br>048-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428698<br>60       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                                                               | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11) : c.12<br>6+186:TMPRSS2_c.39+629<br>23:ERGdel   Protein fusion:<br>in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-003<br>3934-T01-I<br>M6-252 | P-0033<br>934-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428792<br>12       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                               | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+6<br>65:TMPRSS2_c.40-58822:E<br>RGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>2104-T01-I<br>M5-253 | P-0012<br>104-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 428696<br>22       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_001243428) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11): (c.12<br>6+879:TMPRSS2_c.40-630<br>33:ERGdel)   Transcript<br>fusion (TMPRSS2-ERG)       | NA                           |
| EVT-P-001<br>2109-T01-I<br>M5-255 | P-0012<br>109-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428754<br>29       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon1 fused<br>with ERG exons 4-11) : c.55<br>+4448:TMPRSS2_c.40-567<br>00:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                           |
| EVT-P-001<br>2109-T02-I<br>M6-256 | P-0012<br>109-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428754<br>29       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 with<br>ERG exons 4-11) : c.55+444<br>8:TMPRSS2_c.40-56700:ERGdel   Protein fusion: out<br>of frame (TMPRSS2-ERG)             | NA                           |
| EVT-P-001<br>2132-T01-I<br>M5-257 | P-0012<br>132-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428685<br>20       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.126+<br>1526:TMPRSS2_c.18+2872<br>:ERGdel   Protein fusion: in<br>frame (TMPRSS2-ERG)                                                         | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                      | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>2375-T01-I<br>M5-258 | P-0012<br>375-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428701<br>75          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon1 with ERG<br>exons 4-11): c.56-59:TMPR<br>SS2_c.40-58424:ERGdel  <br>Protein fusion: out of frame<br>(TMPRSS2-ERG)   | NA                            |
| EVT-P-001<br>2197-T02-I<br>M5-261 | P-0012<br>197-T0<br>2-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428726<br>50          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2 (NM_001135099<br>- ERG (NM_004449) fusion<br>(TMPRSS2 exon 1 with<br>ERG exons 4-11 ):<br>c.56-2534_c.39+38157del  <br>Protein fusion: out of frame<br>(TMPRSS2-ERG)                | NA                            |
| EVT-P-001<br>2271-T01-I<br>M5-262 | P-0012<br>271-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428728<br>83          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon1 with ERG<br>exons 4-11): c.56-2767:TMP<br>RSS2_c.39+13672:ERGdel<br>  Protein fusion: out of frame<br>(TMPRSS2-ERG) | NA                            |
| EVT-P-003<br>3701-T02-I<br>M6-263 | P-0033<br>701-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428739<br>67          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-3<br>851:TMPRSS2_c.19-2454:E<br>RGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                           | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                        | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>2321-T01-I<br>M5-264 | P-0012<br>321-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868127           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) Fusion (TMPRSS2 exons 1-2 fused with ERG exons 4-11) : c.127-1622:TMPRSS2_c.40-584 17:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-001<br>2327-T01-I<br>M5-266 | P-0012<br>327-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42871029           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exons 2-11) : c.56-913:TMPRSS2_c.-150+107 00:ERGdel   Transcript fusion (TMPRSS2-ERG)           | NA                           |
| EVT-P-001<br>2328-T01-I<br>M5-267 | P-0012<br>328-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42877085           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) Fusion (TMPRSS2 exon 1 fused with ERG exons 2-10) : c.55+2792:TMPRSS2_c.19-630 0:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)  | NA                           |
| EVT-P-001<br>2368-T01-I<br>M5-268 | P-0012<br>368-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869637           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 with ERG exons 2-10): c.126+409:TMPRSS2_c.18+18604:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                         | disrupted_domains | source_annotation_text                                                                                                                                              | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>5650-T01-I<br>M6-269 | P-0025<br>650-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872614           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon 1 with ERG exon 3) : c.56-2498:TMPRSS2_c.39+45031:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>2507-T01-I<br>M5-270 | P-0012<br>507-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872292           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-2176:TMPRSS2_c.18+2263:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                    | NA                           |
| EVT-P-001<br>2555-T01-I<br>M5-271 | P-0012<br>555-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875301           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+4576:TMPRSS2_c.-4791:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                      | NA                           |
| EVT-P-001<br>2555-T02-I<br>M6-272 | P-0012<br>555-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875302           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.55+4576:TMPRSS2_c.40-57551:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>2630-T01-I<br>M5-273 | P-0012<br>630-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 428715<br>97       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 3-11) : c.56<br>-1481:TMPRSS2_c.-47-432<br>6:ERG_del   Transcript<br>fusion (TMPRSS2-ERG)             | NA                           |
| EVT-P-000<br>6856-T01-I<br>M5-276 | P-0006<br>856-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428722<br>07       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11) : c.56<br>-2091:TMPRSS2_c.40-1321<br>3:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                           |
| EVT-P-001<br>8713-T01-I<br>M6-277 | P-0018<br>713-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428679<br>31       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11); c.12<br>7-1426:TMPRSS2_c.40-606<br>31:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG}       | NA                           |
| EVT-P-001<br>2914-T01-I<br>M5-278 | P-0012<br>914-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428691<br>11       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 with<br>ERG exons 2-10); c.126+93<br>5:TMPRSS2_c.18+7096:ER<br>Gdel   Protein fusion: in<br>frame (TMPRSS2-ERG)            | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_nic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                       | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|-------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>2930-T01-I<br>M5-279 | P-0012<br>930-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42870218           | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-102:TMPRSS2_c.18+5504:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                                        | NA                           |
| EVT-P-001<br>3116-T01-I<br>M5-281 | P-0013<br>116-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872374           | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-258:TMPRSS2_c.18+16764:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                                       | NA                           |
| EVT-P-001<br>3336-T01-I<br>M5-282 | P-0013<br>336-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874388           | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused with ERG exons 2 to 10) : c.56-4272:TMPRSS2_c.19-13796:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-001<br>3372-T01-I<br>M5-283 | P-0013<br>372-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42868312           | 2               | 42                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.126+1734:TMPRSS2_c.237-2384:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                                     | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>3401-T01-I<br>M5-284 | P-0013<br>401-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875769           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+4108:TMPRSS2_c.19-22056:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                            | NA                           |
| EVT-P-001<br>3457-T01-I<br>M5-285 | P-0013<br>457-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | three_prime | 42864658           | 3               | 117                         | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.5:ERG_c.349+1625:TMPRSS2dup   Protein fusion: mid-exon (ERG-TMPRSS2)                                               | NA                           |
| EVT-P-001<br>3465-T01-I<br>M5-286 | P-0013<br>465-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869204           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.126+842:TMPRSS2_c.40-57378:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)                                                | NA                           |
| EVT-P-001<br>3503-T01-I<br>M5-287 | P-0013<br>503-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875717           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10); c.55+4160:TMPRSS2_c.18+11744:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>3226-T01-I<br>M6-288 | P-0033<br>226-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871204           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon1 fused with ERG exon2) : c.56-1088:TMPRSS2_c.40-47540:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}              | NA                           |
| EVT-P-002<br>6159-T01-I<br>M6-289 | P-0026<br>159-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42877818           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon1 fused to ERG exon2) : c.55+2059:TMPRSS2_c.19-16879:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                | NA                           |
| EVT-P-001<br>3729-T01-I<br>M5-290 | P-0013<br>729-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876881           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10); c.55+2996:TMPRSS2_c.18+1445:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)          | NA                           |
| EVT-P-001<br>3733-T01-I<br>M5-291 | P-0013<br>733-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868379           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in-frame with ERG exons 2-10); c.126+1667:TMPRSS2_c.18+2781:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                              | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0013791-T01-I M5-292 | P-0013791-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42869530           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exons 1-2 fused with ERG exons 4-11) : c.126+516:TMPRSS2_c.40-64952:ERGdel   Protein fusion: in frame (TMPRSS2-ERG)        | NA                           |
| EVT-P-0013810-T01-I M5-293 | P-0013810-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42867882           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused in-frame to ERG exons 4-11): c.127-1377:TMPRSS2_c.40-64863:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-0013837-T01-I M5-294 | P-0013837-T01-IM5 |            | ERG-TM PRSS2 | ERG          | unknown      | five_prime  | 42875045           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 3-11): c.55+4832:TMPRSS2_c.-149-29687:ERGdel   Transcript fusion (TMPRSS2-ERG)                   | NA                           |
| EVT-P-0013844-T01-I M5-295 | P-0013844-T01-IM5 |            | ERG-TM PRSS2 | ERG          | in-frame     | three_prime | 42867863           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_004449) - TMPRSS2 (NM_001135099) rearrangement: c.40-54335:ERG_c.127-1358:TMPRSS2dup   Protein fusion: in frame (ERG-TMPRSS2)                                               | NA                           |
| EVT-P-0013844-T02-I M6-296 | P-0013844-T02-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | three_prime | 42867863           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_004449) - TMPRSS2 (NM_001135099) rearrangement: c.40-54335:ERG_c.127-1358:TMPRSS2dup   Protein fusion: in frame (ERG-TMPRSS2)                                               | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status    | tar_get_role    | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_domains | lost_do_mains                                                                             | disrupt_ed_domains | source_annotation_text                                                                                                                                                                                | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|------------------|-----------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------|-------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>3858-T01-I<br>M5-297 | P-0013<br>858-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428755<br>33          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11) c.55+<br>4344:TMPRSS2_c.39+4939<br>1:ERGdel   Protein fusion:<br>out of frame<br>(TMPRSS2-ERG) | NA                            |
| EVT-P-001<br>3957-T01-I<br>M5-300 | P-0013<br>957-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428752<br>89          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+4<br>588:TMPRSS2_c.40-54610:<br>ERGdel   Protein fusion: out<br>of frame (TMPRSS2-ERG)                                                    | NA                            |
| EVT-P-001<br>4004-T01-I<br>M5-301 | P-0014<br>004-T0<br>1-IM5 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428760<br>49          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion:<br>c.55+3828_c.40-59410del  <br>Protein fusion: out of frame<br>(TMPRSS2-ERG)                                                                  | NA                            |
| EVT-P-001<br>4004-T02-I<br>M6-302 | P-0014<br>004-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428760<br>49          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion:<br>c.55+3828_c.40-59410del  <br>Protein fusion: out of frame<br>(TMPRSS2-ERG)                                                                  | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4004-T03-I<br>M6-303 | P-0014<br>004-T0<br>3-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876050           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.55+3827_c.40-59410del   Protein Fusion: out of frame {TMPRSS2:ERG}                                                           | NA                           |
| EVT-P-001<br>4076-T01-I<br>M5-304 | P-0014<br>076-T0<br>1-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873749           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-3633:TMPRSS2_c.18+24406:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)      | NA                           |
| EVT-P-001<br>4202-T01-I<br>M6-306 | P-0014<br>202-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869590           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in-frame to ERG exons 2-10): c.126+456:TMPRSS2_c.18+59:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |
| EVT-P-002<br>6162-T01-I<br>M6-307 | P-0026<br>162-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872550           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon 1 with ERG exons 4-10): c.56-2434:TMPRSS2_c.40-58863:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}          | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                        | lost_domains                                                                                                                                                      | disrupted_domains | source_annotation_text                                                                                                                                                                             | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4341-T01-I<br>M6-308 | P-0014<br>341-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428723<br>63       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                                         | Trypsin; Sca<br>ven<br>ger<br>r<br>e<br>c<br>e<br>p<br>t<br>o<br>r<br>c<br>y<br>s<br>t<br>e<br>i<br>n<br>e<br>-<br>r<br>i<br>c<br>h<br>d<br>o<br>m<br>a<br>i<br>n |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-10); c.56-224<br>7:TMPRSS2_c.-4097:ERGd<br>el   Protein fusion: out of<br>frame (TMPRSS2-ERG)     | NA                           |
| EVT-P-003<br>2801-T01-I<br>M6-309 | P-0032<br>801-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428707<br>11       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                                         | Trypsin; Sca<br>ven<br>ger<br>r<br>e<br>c<br>e<br>p<br>t<br>o<br>r<br>c<br>y<br>s<br>t<br>e<br>i<br>n<br>e<br>-<br>r<br>i<br>c<br>h<br>d<br>o<br>m<br>a<br>i<br>n |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 2-10); c.56-<br>595:TMPRSS2_c.19-12368:<br>ERGdel   Protein Fusion: out<br>of frame (TMPRSS2:ERG) | NA                           |
| EVT-P-003<br>2799-T01-I<br>M6-311 | P-0032<br>799-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428790<br>62       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                                         | Trypsin; Sca<br>ven<br>ger<br>r<br>e<br>c<br>e<br>p<br>t<br>o<br>r<br>c<br>y<br>s<br>t<br>e<br>i<br>n<br>e<br>-<br>r<br>i<br>c<br>h<br>d<br>o<br>m<br>a<br>i<br>n |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11); c.55+815:<br>TMPRSS2_c.39+61431:ER<br>Gdel   Protein Fusion: out of<br>frame (TMPRSS2:ERG)   | NA                           |
| EVT-P-001<br>4414-T01-I<br>M6-312 | P-0014<br>414-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | three_prime | 428725<br>04       | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Sca<br>venger<br>recept<br>or cyst<br>eine-ri<br>ch do<br>main |                                                                                                                                                                   |                   | ERG (NM_004449) -<br>TMPRSS2<br>(NM_001135099)<br>rearrangement: c.39+51460:<br>ERG_c.56-2388:TMPRSS2<br>dup   Protein fusion: out of<br>frame (ERG-TMPRSS2)                                       | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                         | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4425-T01-I<br>M6-313 | P-0014<br>425-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872644           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-2528:TMPRSS2_c.19-4354: ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)   | NA                           |
| EVT-P-001<br>4438-T01-I<br>M6-314 | P-0014<br>438-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875935           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-10): c.55+3942:TMPRSS2_c.39+50799: ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)    | NA                           |
| EVT-P-002<br>6196-T01-I<br>M6-315 | P-0026<br>196-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873713           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.126+749:TMPRSS2_c.40-60753: ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-000<br>0373-T02-I<br>M5-316 | P-0000<br>373-T0<br>2-IM5 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874110           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-3994:TMPRSS2_c.19-24499: ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                             | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4531-T01-I<br>M6-317 | P-0014<br>531-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874070           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) Fusion: c.56-3954:TMPRSS2_c.18+11943:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)                                            | NA                           |
| EVT-P-003<br>2663-T01-I<br>M6-318 | P-0032<br>663-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42877555           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.55+2322:TMPRSS2_c.18+12726:ERGdel   Protein Fusion: out of frame (TMPRSS2:ERG)  | NA                           |
| EVT-P-003<br>2646-T01-I<br>M6-319 | P-0032<br>646-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872447           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1 fused to ERG exons 2-10): c.56-2331:TMPRSS2_c.18+11761:ERGdel   Protein Fusion: out of frame (TMPRSS2:ERG) | NA                           |
| EVT-P-003<br>2573-T01-I<br>M6-320 | P-0032<br>573-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868708           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) Fusion (TMPRSS2 exon2 fused with ERG exon4: c.126+133:TMPRSS2_c.40-60824:ERGdel   Protein Fusion: in frame (TMPRSS2:ERG)            | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4841-T01-I<br>M6-321 | P-0014<br>841-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873166           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon2 fused to ERG exon 4) : c.56-3050:TMPRSS2_c.39+61126:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)      | NA                           |
| EVT-P-001<br>4992-T01-I<br>M6-323 | P-0014<br>992-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42877448           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) Fusion (TMPRSS2 exon 1 fused with ERG exons 2-10) : c.55+2429:TMPRSS2_c.18+5301:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG) | NA                           |
| EVT-P-003<br>2046-T01-I<br>M6-324 | P-0032<br>046-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875015           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.55+4862:TMPRSS2_c.40-58959:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                           | NA                           |
| EVT-P-001<br>5015-T01-I<br>M6-325 | P-0015<br>015-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868021           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11) : c.127-1516:TMPRSS2_c.40-44007:ERGdel   Protein fusion: in frame (TMPRSS2-ERG) | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_d_o_mains | lost_d_o_mains                                                                               | disrupt_ed_d_o_mains | source_annotation_text                                                                                                                                                                  | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------|----------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>2027-T01-I<br>M6-326 | P-0032<br>027-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | unk<br>now<br>n      | fiv<br>e_<br>pri<br>me | 428663<br>29          | 3                | 101                          | False                    | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon3 with ERG<br>exon2) : c.303:TMPRSS2_c.<br>19-1548:ERGdel   Protein<br>Fusion: mid-exon<br>{TMPRSS2:ERG}          | NA                            |
| EVT-P-003<br>1719-T01-I<br>M6-327 | P-0031<br>719-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | unk<br>now<br>n      | fiv<br>e_<br>pri<br>me | 428713<br>55          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG(NM_182918) Fusion<br>(TMPRSS2 exon2 fused<br>with ERG exon1 ) : c.56-123<br>9.3:TMPRSS2_c.-901:ERG<br>del   Transcript Fusion<br>{TMPRSS2:ERG}          | NA                            |
| EVT-P-003<br>1599-T01-I<br>M6-328 | P-0031<br>599-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428757<br>15          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.55+4<br>162:TMPRSS2_c.19-26160:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                      | NA                            |
| EVT-P-002<br>4679-T01-I<br>M6-329 | P-0024<br>679-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | unk<br>now<br>n      | fiv<br>e_<br>pri<br>me | 428748<br>63          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                     | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                      | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exon 4-11 ) : c.56-474<br>7:TMPRSS2_c.40-54816:E<br>RGdel   Transcript Fusion<br>{TMPRSS2:ERG} | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                            | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>1536-T01-I<br>M6-330 | P-0031<br>536-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428690<br>29       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.126+<br>1017:TMPRSS2_c.18+1301<br>4:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG}                                                   | NA                           |
| EVT-P-001<br>5132-T01-I<br>M6-332 | P-0015<br>132-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428671<br>30       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 4-11):c.127<br>-625:TMPRSS2_c.40-56456<br>:ERGdel   Protein fusion: in<br>frame (TMPRSS2-ERG) | NA                           |
| EVT-P-001<br>5209-T01-I<br>M6-333 | P-0015<br>209-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428776<br>85       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon1 fused to<br>ERG exon2): c.55+2192:TM<br>PRSS2_c.18+13090:ERGdel<br>  Protein fusion: out of<br>frame (TMPRSS2-ERG)        | NA                           |
| EVT-P-001<br>5250-T01-I<br>M6-334 | P-0015<br>250-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428744<br>26       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exon 2): c.56-4310:TM<br>PRSS2_c.19-20348:ERGdel<br>  Protein fusion: out of frame<br>(TMPRSS2-ERG)      | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>1429-T01-I<br>M6-335 | P-0031<br>429-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872684           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon 1 fused to ERG exon 2-10) : c.56-2568:TMPRSS2_c.40-46240:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-003<br>1412-T01-I<br>M6-336 | P-0031<br>412-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876045           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon2 fused with ERG exon2) : c.55+3832:TMPRSS2_c.19-22354:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}    | NA                           |
| EVT-P-003<br>1374-T01-I<br>M6-338 | P-0031<br>374-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873062           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.56-2946:TMPRSS2_c.40-59140:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                          | NA                           |
| EVT-P-001<br>5482-T01-I<br>M6-339 | P-0015<br>482-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871231           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon1 fused to ERG exon 2): c.56-1115:TMPRSS2_c.18+11715:ERGdel   Protein fusion: out of frame (TMPRSS2-ERG)      | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status    | tar_get_role    | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_doma_in | lost_doma_in                                                                                 | disrupt_ed_doma_in | source_annotation_text                                                                                                                                                                           | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|------------------|-----------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------|----------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>5628-T01-I<br>M6-340 | P-0015<br>628-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428701<br>43          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10); c.56-27:T<br>MPSS2_c.18+6821:ERGdel<br>  Protein fusion: out of frame<br>(TMPRSS2-ERG)     | NA                            |
| EVT-P-001<br>5744-T01-I<br>M6-342 | P-0015<br>744-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428710<br>33          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11); c.56-917:<br>TMPRSS2_c.39+7155:ERG<br>del   Protein fusion: out of<br>frame (TMPRSS2-ERG)  | NA                            |
| EVT-P-001<br>5744-T02-I<br>M6-343 | P-0015<br>744-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428710<br>34          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11); c.56-918:<br>TMPRSS2_c.39+7155:ERG<br>del   Protein Fusion: out of<br>frame {TMPRSS2:ERG}  | NA                            |
| EVT-P-001<br>5837-T01-I<br>M6-344 | P-0015<br>837-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428714<br>40          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10); c.56-132<br>4:TMPSS2_c.18+10744:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG} | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0015850-T01-IM6-345 | P-0015850-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42878843           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 to ERG exon 4): c.55+1034:TMPRSS2_c.39+9192:ERG   Protein Fusion: out of frame {TMPRSS2:ERG}             | NA                           |
| EVT-P-0031269-T01-IM6-346 | P-0031269-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42870853           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) Fusion (TMPRSS2 exon2 fused with ERG exon2) : c.56-737:TMPRSS2_c.19-13490:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}    | NA                           |
| EVT-P-0015960-T01-IM6-347 | P-0015960-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42870232           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exon 4): c.56-116:TMPRSS2_c.40-64777:ERG   Protein Fusion: out of frame {TMPRSS2:ERG}       | NA                           |
| EVT-P-0015960-T02-IM6-348 | P-0015960-T02-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42870232           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11):c.56-116:TMPRSS2_c.40-64777:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                        | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6015-T01-I<br>M6-349 | P-0016<br>015-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428706<br>64       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exon 4) : c.56-548:TM<br>PRSS2_c.40-58734:ERGdel<br>  Protein Fusion: out of<br>frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-001<br>6063-T01-I<br>M6-350 | P-0016<br>063-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428670<br>18       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1-2 with<br>ERG exon 4-11) : c.127-513<br>:TMPRSS2_c.40-57914:ER<br>Gdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}    | NA                           |
| EVT-P-001<br>6087-T01-I<br>M6-351 | P-0016<br>087-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428740<br>68       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 with<br>ERG exons 4-11) : c.56-395<br>2:TMPRSS2_c.39+61847:ER<br>Gdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>6134-T01-I<br>M6-352 | P-0016<br>134-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428705<br>09       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 with<br>ERG exons 2-10) : c.56-393<br>:TMPRSS2_c.18+26221:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}  | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6236-T01-I<br>M6-354 | P-0016<br>236-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872990           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exon 4): c.56-2874:TMPRSS2_c.39+13664:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}       | NA                           |
| EVT-P-001<br>6296-T01-I<br>M6-355 | P-0016<br>296-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867024           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) fusion (TMPRSS2 exon 1 with ERG exons 2-10) : c.40-46781:TMPRSS2_c.18+5962:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}           | NA                           |
| EVT-P-001<br>6325-T01-I<br>M6-357 | P-0016<br>325-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42868979           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.126+1067:TMPRSS2_c.40-55287:ERGdel   Transcript Fusion {TMPRSS2:ERG}          | NA                           |
| EVT-P-003<br>1178-T02-I<br>M6-358 | P-0031<br>178-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874466           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-4350:TMPRSS2_c.18+11076:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                             | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                               | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|-------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>1178-T01-I<br>M6-359 | P-0031<br>178-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428744<br>66          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-4<br>350:TMPRSS2_c.18+11076<br>:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}                                                | NA                            |
| EVT-P-001<br>6396-T01-I<br>M6-360 | P-0016<br>396-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428692<br>49          | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1-2 with<br>ERG exons 2-10) : c.126+79<br>7:TMPRSS2_c.18+1126:ER<br>Gdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}          | NA                            |
| EVT-P-003<br>0984-T01-I<br>M6-363 | P-0030<br>984-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428748<br>57          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.56-4<br>741:TMPRSS2_c.40-57465:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                   | NA                            |
| EVT-P-002<br>6212-T01-I<br>M6-364 | P-0026<br>212-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428738<br>52          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 4-11): c.126+<br>749:TMPRSS2_c.40-60753:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG} | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6698-T01-I<br>M6-365 | P-0016<br>698-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872715           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10); c.56-259:TMPRSS2_c.18+9969:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}        | NA                           |
| EVT-P-001<br>6733-T01-I<br>M6-366 | P-0016<br>733-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873910           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG fusion (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10); c.56-379:TMPRSS2_c.19-6151:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-003<br>0872-T01-I<br>M6-368 | P-0030<br>872-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872678           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-2562:TMPRSS2_c.18+12161:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                               | NA                           |
| EVT-P-003<br>0845-T01-I<br>M6-369 | P-0030<br>845-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42870695           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon 1 with ERG exons 4-10); c.56-579:TMPRSS2_c.40-59079:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}           | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>0812-T01-I<br>M6-370 | P-0030<br>812-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867559           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.127-1054:TMPRSS2_c.40-5688<br>0:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG}                                                           | NA                           |
| EVT-P-001<br>6895-T01-I<br>M6-372 | P-0016<br>895-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873673           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 2-10) : c.56-3557:TMPRSS2_c.19-1378<br>3:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>6933-T01-I<br>M6-373 | P-0016<br>933-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875065           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon1 fused<br>with ERG exons 4-11) : c.55+4812:TMPRSS2_c.40-588<br>94:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}     | NA                           |
| EVT-P-001<br>6983-T01-I<br>M6-374 | P-0016<br>983-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867872           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>fusion (NM_182918) fusion<br>(TMPRSS2 exon 1-2 fused<br>to ERG exons 2-10): c.127-1367:TMPRSS2_c.19-1543<br>1:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG}   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>7065-T01-I<br>M6-376 | P-0017<br>065-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871083           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG Fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.56-967:TMPRSS2_c.40-42355:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}            | NA                            |
| EVT-P-001<br>7168-T01-I<br>M6-377 | P-0017<br>168-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868343           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_182918) fusion (TMPRSS2 exon 1-2 with ERG exons 2-10) : c.126+1703:TMPRSS2_c.19-17382:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}      | NA                            |
| EVT-P-001<br>7235-T01-I<br>M6-378 | P-0017<br>235-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874411           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) Fusion (TMPRSS2 exon 1-2 with ERG exon 4-11) : c.56-4295:TMPRSS2_c.39+62371:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}    | NA                            |
| EVT-P-002<br>6409-T01-I<br>M6-379 | P-0026<br>409-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42878494           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.55+1383:TMPRSS2_c.18+21976:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                            |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0033300-T01-I M6-381 | P-0033300-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42878752           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+125:TMPRSS2_c.19-12860:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                | NA                           |
| EVT-P-0017376-T01-I M6-382 | P-0017376-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872320           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11); c.56-2204:TMPRSS2_c.40-61559:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}      | NA                           |
| EVT-P-0026228-T01-I M6-384 | P-0026228-T01-IM6 |            | ERG-TM PRSS2 | ERG          | in-frame     | five_prime  | 42868990           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in-frame to ERG exons 2-10); c.126+1056:TMPRSS2_c.18+9:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                           |
| EVT-P-0017548-T01-I M5-385 | P-0017548-T01-IM5 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872954           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11); c.56-2838:TMPRSS2_c.40-58598:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}      | NA                           |

| event_id                   | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0017548-T02-I M6-386 | P-0017548-T02-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42872954           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11): c.56-2838:TMPRSS2_c.40-58598:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}   | NA                           |
| EVT-P-0017708-T01-I M6-387 | P-0017708-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42874792           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exons 4-11): c.56-4676:TMPRSS2_c.40-56300:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-0017730-T01-I M6-388 | P-0017730-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42874582           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-4466:TMPRSS2_c.19-8247:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}    | NA                           |
| EVT-P-0024488-T01-I M6-389 | P-0024488-T01-IM6 |            | ERG-TM PRSS2 | ERG          | out-of-frame | five_prime  | 42876909           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon1 fused with ERG exons 4-11) : c.55+2968:TMPRSS2_c.39+16108:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_ge_ne | fra_me_st_atu_s      | tar_get_r_ole          | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                             | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|----------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|--------------------|-------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>0675-T01-I<br>M6-390 | P-0030<br>675-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428779<br>73          | 1                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 5-11); c.55+190<br>4:TMPRSS2_c.40-59905:E<br>RGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG} | NA                            |
| EVT-P-003<br>0580-T01-I<br>M6-391 | P-0030<br>580-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428731<br>12          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-2<br>996:TMPRSS2_c.18+1813:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                 | NA                            |
| EVT-P-003<br>0465-T01-I<br>M6-392 | P-0030<br>465-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428728<br>82          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10); c.56-276<br>6:TMPRSS2_c.19-5647:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}  | NA                            |
| EVT-P-003<br>0381-T01-I<br>M6-393 | P-0030<br>381-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428760<br>14          | 1                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion : c.55+<br>3863:TMPRSS2_c.18+9727<br>:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}                                             | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                                      | disrupted_domains | source_annotation_text                                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|-----------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>8038-T01-I<br>M6-394 | P-0018<br>038-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873173           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion : c.56-3057:TMPRSS2_c.18+1102<br>1:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}                                     | NA                           |
| EVT-P-001<br>8090-T01-I<br>M6-395 | P-0018<br>090-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867125           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10):c.127-620:TMPRSS2_c.19-2241:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG} | NA                           |
| EVT-P-003<br>0261-T01-I<br>M6-396 | P-0030<br>261-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875512           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+4365:TMPRSS2_c.39+58079<br>:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}                                      | NA                           |
| EVT-P-001<br>8149-T01-I<br>M6-398 | P-0018<br>149-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876321           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TMPRSS2 exon 1 with<br>ERG exon 4) : c.55+3556:TMPRSS2_c.40-56840:ERGdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}      | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status    | tar_get_role    | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_doma_in | lost_doma_in                                                                                 | disrupt_ed_doma_in | source_annotation_text                                                                                                                                                                                         | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|------------------|-----------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------|----------------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>8315-T01-I<br>M6-400 | P-0018<br>315-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame     | fiv_e_pri<br>me | 428684<br>71          | 2                | 42                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>in-frame with ERG exons<br>2-10): c.126+1575:TMPRSS<br>2_c.18+18761:ERGdel  <br>Protein Fusion: in frame<br>{TMPRSS2:ERG} | NA                            |
| EVT-P-001<br>8395-T01-I<br>M6-401 | P-0018<br>395-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | unk<br>now<br>n  | fiv_e_pri<br>me | 428774<br>13          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+2<br>464:TMPRSS2_c.40-54367:<br>ERGdel   Transcript Fusion<br>{TMPRSS2:ERG}                                                                        | NA                            |
| EVT-P-001<br>8489-T01-I<br>M6-402 | P-0018<br>489-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428739<br>76          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) -ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 2-11): c.56-<br>3860:TMPRSS2_c.40-6101:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}              | NA                            |
| EVT-P-003<br>0187-T01-I<br>M6-403 | P-0030<br>187-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | out-of-fr<br>ame | fiv_e_pri<br>me | 428746<br>62          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.56-4<br>546:TMPRSS2_c.18+5296:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                              | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_nic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                | disrupted_domains | source_annotation_text                                                                                                                                                                                   | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|-------------------------|---------------|--------------------------|---------------------|------------------|-------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>8581-T01-I<br>M6-404 | P-0018<br>581-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428680<br>59       | 3               | 43                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10): c.127-<br>1554:TMPRSS2_c.18+8829<br>:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}         | NA                           |
| EVT-P-001<br>8704-T01-I<br>M6-405 | P-0018<br>704-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428724<br>58       | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11): c.56-<br>2342:TMPRSS2_c.39+6367<br>8:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG}   | NA                           |
| EVT-P-002<br>5650-T02-I<br>M6-406 | P-0025<br>650-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428726<br>14       | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11):c.56-2498<br>:TMPRSS2_c.39+45031:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}         | NA                           |
| EVT-P-002<br>5455-T01-I<br>M6-407 | P-0025<br>455-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428722<br>83       | 2               | 19                          | True                    | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 2-10): c.56-<br>2167:TMPRSS2_c.18+1698<br>:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                         | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>5571-T01-I<br>M6-408 | P-0025<br>571-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872864           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-2748:TMPRSS2_c.18+2970:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}   | NA                            |
| EVT-P-001<br>8886-T01-I<br>M6-410 | P-0018<br>886-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872880           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-2764:TMPRSS2_c.18+8733:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                            |
| EVT-P-003<br>0109-T01-I<br>M6-411 | P-0030<br>109-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867208           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.56-2838:TMPRSS2_c.40-58598:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}                                               | NA                            |
| EVT-P-002<br>4461-T01-I<br>M6-412 | P-0024<br>461-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872210           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449): c.56-2094:TMPRSS2_c.40-59950:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}                                                  | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>9212-T01-I<br>M6-413 | P-0019<br>212-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876132           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exon 2): c.55+3745:TMPRSS2_c.19-4616:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>9212-T02-I<br>M6-414 | P-0019<br>212-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876132           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exon 2): c.55+3745:TMPRSS2_c.19-4616:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>9212-T03-I<br>M6-415 | P-0019<br>212-T0<br>3-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42876132           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exon 2): c.55+3745:TMPRSS2_c.19-4616:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-003<br>0086-T01-I<br>M6-416 | P-0030<br>086-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | three_prime | 42866765           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.19-5914:ERG_c.127-260:TMPRSS2dup   Protein Fusion: in frame {ERG:TMPRSS2}                                   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                           | disrupted_domains | source_annotation_text                                                                                                                                                    | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>9347-T01-I<br>M6-417 | P-0019<br>347-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428725<br>35       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-2419:TMPRSS2_c.19-7767:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>9692-T01-I<br>M6-419 | P-0019<br>692-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428782<br>73       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11):c.55+1604:TMPRSS2_c.40-60431:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-001<br>9695-T01-I<br>M6-420 | P-0019<br>695-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428710<br>57       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11):c.56-941:TMPRSS2_c.40-61030:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-001<br>9797-T01-I<br>M6-421 | P-0019<br>797-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428722<br>15       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca ven ger r ece ptor cyst eine -rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-2099:TMPRSS2_c.19-24707:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}     | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_ge_ne | fra_me_st_atu_s      | tar_get_r_ole          | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                                  | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|----------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>9840-T01-I<br>M6-422 | P-0019<br>840-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428742<br>33          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) Fusion<br>(TMPRSS2 exon1 with ERG<br>exons 2-10) : c.56-4117:TM<br>PRSS2_c.18+11302:ERGde<br>l   Protein Fusion: out of<br>frame {TMPRSS2:ERG}           | NA                            |
| EVT-P-002<br>9939-T01-I<br>M6-423 | P-0029<br>939-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428750<br>25          | 1                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+4<br>852:TMPRSS2_c.40-56952:<br>ERG   Protein Fusion: out of<br>frame {TMPRSS2:ERG}                                                         | NA                            |
| EVT-P-001<br>9930-T01-I<br>M6-424 | P-0019<br>930-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428709<br>42          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11): c.56-<br>826:TMPRSS2_c.40-59325:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}      | NA                            |
| EVT-P-001<br>9933-T01-I<br>M6-425 | P-0019<br>933-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G        | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428748<br>08          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion (<br>TMPRSS2 exon 1 fused<br>with ERG exons 4-11): c.56-<br>4692:TMPRSS2_c.39+6002<br>1:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG} | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>6238-T01-I<br>M6-426 | P-0026<br>238-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42875622           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.55+4255:TMPRSS2_c.18+13682:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>0078-T01-I<br>M6-427 | P-0020<br>078-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872724           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-2608:TMPRSS2_c.236+3010:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}      | NA                           |
| EVT-P-002<br>0083-T01-I<br>M6-428 | P-0020<br>083-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873200           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) (TMPRSS2 exon 1 fused with ERG exons 2-10): c.56-3084:TMPRSS2_c.18+8091:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}       | NA                           |
| EVT-P-002<br>0132-T01-I<br>M6-430 | P-0020<br>132-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867623           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11):c.127-1118:TMPRSS2_c.39+56940:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}  | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-000<br>0373-T01-I<br>M3-431 | P-0000<br>373-T0<br>1-IM3 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874110           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) c.56-3994_c.19-24499del   Protein fusion: out of frame (TMPRSS2-ERG)                                                              | NA                           |
| EVT-P-002<br>9811-T01-I<br>M6-432 | P-0029<br>811-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871857           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 2 fused to ERG exons 2-10): c.56-1741:TMPRSS2_c.18+17833:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>0424-T01-I<br>M6-434 | P-0020<br>424-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42879287           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11):c.55+590:TMPRSS2_c.40-60250:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}   | NA                           |
| EVT-P-002<br>0750-T01-I<br>M6-435 | P-0020<br>750-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | three_prime | 42867090           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_004449) - TMPRSS2 (NM_001135099) rearrangement: c.39+2008:ERG_c.127-585:TMPRSS2 dup   Protein Fusion: in frame {ERG:TMPRSS2}                                       | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>9809-T01-I<br>M6-437 | P-0029<br>809-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428684<br>49       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1-2 with<br>ERG exons 4-10) : c.126+15<br>97:TMPRSS2_c.39+39940:<br>ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}          | NA                           |
| EVT-P-002<br>0812-T01-I<br>M6-438 | P-0020<br>812-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428762<br>39       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) (TMPRSS2<br>exon 1 fused with ERG<br>exons 4-11): c.55+3638:TM<br>PRSS2_c.39+63429:ERGdel<br>  Protein Fusion: out of<br>frame {TMPRSS2:ERG}            | NA                           |
| EVT-P-002<br>0812-T02-I<br>M6-439 | P-0020<br>812-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428762<br>39       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused<br>with ERG exons 4-11): c.55<br>+3638:TMPRSS2_c.39+634<br>29:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>9796-T01-I<br>M6-440 | P-0029<br>796-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428763<br>88       | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.55+3<br>489:TMPRSS2_c.40-58375:<br>ERGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}                                                     | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status   | tar_get_role    | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                             | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|-----------------|-----------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>0844-T01-I<br>M6-441 | P-0020<br>844-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame    | fiv_e_pri<br>me | 428681<br>31          | 3                | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 4-11):c.127-1<br>626:TMPRSS2_c.40-61304:<br>ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}   | NA                            |
| EVT-P-002<br>0847-T02-I<br>M6-442 | P-0020<br>847-T0<br>2-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame    | fiv_e_pri<br>me | 428671<br>06          | 3                | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 2-10): c.12<br>7-601:TMPRSS2_c.19-1735<br>7:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG} | NA                            |
| EVT-P-002<br>9405-T01-I<br>M6-443 | P-0029<br>405-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | in-fr<br>ame    | fiv_e_pri<br>me | 428679<br>32          | 3                | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10):c.127-1<br>427:TMPRSS2_c.18+20762<br>:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}   | NA                            |
| EVT-P-002<br>1104-T01-I<br>M6-444 | P-0021<br>104-T0<br>1-IM6 |              | ERG-TM<br>PRSS2 | ER<br>G       | unk<br>now<br>n | fiv_e_pri<br>me | 428665<br>60          | 3                | 43                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 4-11):c.127-5<br>5:TMPRSS2_c.40-55445:E<br>RGdel   Transcript Fusion<br>{TMPRSS2:ERG}            | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                             | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>9333-T01-I<br>M6-445 | P-0029<br>333-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428739<br>97       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1 - ERG<br>exons 4-11): c.56-3881:TMP<br>RSS2_c.40-57209:ERG  <br>Protein Fusion: out of frame<br>{TMPRSS2:ERG}            | NA                           |
| EVT-P-002<br>1183-T01-I<br>M6-446 | P-0021<br>183-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428693<br>87       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>with ERG exons 2-10): c.12<br>6+659:TMPRSS2_c.18+256<br>15:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>1239-T01-I<br>M6-447 | P-0021<br>239-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428735<br>52       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11):c.56-3436<br>:TMPRSS2_c.40-63140:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}   | NA                           |
| EVT-P-002<br>4249-T01-I<br>M6-448 | P-0024<br>249-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428707<br>24       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-rich<br>domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion (<br>TMPRSS2 exon 1 fused to<br>ERG exons 4 -11): c.56-608<br>:TMPRSS2_c.39+56905:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>9163-T01-I<br>M6-449 | P-0029<br>163-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42870139           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.56-23:TMPRSS2_c.39+33015:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>9105-T02-I<br>M6-450 | P-0029<br>105-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873116           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-3000:TMPRSS2_c.18+16661:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>9105-T01-I<br>M6-451 | P-0029<br>105-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873116           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10): c.56-3000:TMPRSS2_c.18+16661:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>1310-T01-I<br>M6-452 | P-0021<br>310-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | three_prime | 42868727           | 2               | 42                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_004449) - TMPRSS2 (NM_001135099) rearrangement: c.39+53911:ERG_c.126+1319:TMPRSS2dup   Protein Fusion: in frame {ERG:TMPRSS2}                                       | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                        | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>1342-T01-I<br>M6-453 | P-0021<br>342-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868257           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.127-1752:TMPRSS2_c.39+5169 6:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}   | NA                           |
| EVT-P-002<br>1388-T01-I<br>M6-454 | P-0021<br>388-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42866534           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.127-29:TMPRSS2_c.39+17938: ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}     | NA                           |
| EVT-P-002<br>4217-T02-I<br>M6-455 | P-0024<br>217-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874589           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004444) fusion (TMPRSS2 exons 1-2 fused with ERG exons 4-10): c.56-4473:TMPRSS2_c.39+1935 1:ERG   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>8587-T01-I<br>M6-456 | P-0028<br>587-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871201           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10): c.-56-1085:TMPRSS2_c.18+655 1:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                  | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|---------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>1577-T01-I<br>M6-457 | P-0021<br>577-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428694<br>77       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10): c.126+<br>569:TMPRSS2_c.19-13687:<br>ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}       | NA                           |
| EVT-P-002<br>1673-T01-I<br>M6-458 | P-0021<br>673-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428740<br>55       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10): c.56-393<br>9:TMPRSS2_c.18+2925:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}       | NA                           |
| EVT-P-002<br>8569-T01-I<br>M6-459 | P-0028<br>569-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428711<br>44       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 2 fused<br>with ERG exons 4-10): c.56-<br>1028:TMPRSS2_c.39+4079<br>9:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>1698-T01-I<br>M6-460 | P-0021<br>698-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428725<br>83       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenger<br>receptor<br>cysteine-<br>rich domain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10): c.56-2<br>467:TMPRSS2_c.19-2116:E<br>RGdel   Protein Fusion: out<br>of frame {TMPRSS2:ERG}    | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>8551-T01-I<br>M6-461 | P-0028<br>551-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42872253           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon 1 fused to ERG exons 4-11); c.56-2137:TMPRSS_c.40-54306:ERGdel   Transcript Fusion {TMPRSS2:ERG}          | NA                           |
| EVT-P-002<br>1853-T01-I<br>M6-462 | P-0021<br>853-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873148           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) -ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exon4) : c.56-3032:TMPRSS2_c.40-59421:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>2060-T01-I<br>M6-463 | P-0022<br>060-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | three_prime | 42861857           | 4               | 117                         | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.236+78:ERG_c.350-337:TMPRSS2dup   Protein Fusion: out of frame {ERG:TMPRSS2}                                  | NA                           |
| EVT-P-002<br>2117-T01-I<br>M6-464 | P-0022<br>117-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42866328           | 3               | 102                         | False                  | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-3 fused to ERG exons 2-10); c.304:TMPRSS2_c.18+8044:ERGdel   Protein Fusion: mid-exon {TMPRSS2:ERG}    | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>8430-T01-I<br>M6-465 | P-0028<br>430-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873954           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon 1 with ERG exons 4-10).   Protein Fusion: out of frame {TMPRSS2:ERG}                                         | NA                           |
| EVT-P-002<br>4217-T01-I<br>M6-467 | P-0024<br>217-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874589           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11):c.56-4473:TMPRSS2_c.39+19351:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>7526-T01-I<br>M6-470 | P-0027<br>526-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869752           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449.4) fusion (TMPRSS2 exon 1-2 with ERG exons 4-10) : c.126+294:TMPRSS2_c.40-60810:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}    | NA                           |
| EVT-P-002<br>2446-T01-I<br>M6-471 | P-0022<br>446-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868538           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11): c.126+1508:TMPRSS2_c.39+58986:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                           | disrupted_domains | source_annotation_text                                                                                                                                                                            | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>2470-T01-I<br>M6-472 | P-0022<br>470-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428727<br>73       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven ger r<br>ece ptor<br>cyst eine<br>-rich dom<br>ain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 4-11):c.56-2657<br>:TMPRSS2_c.40-63048:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>7454-T01-I<br>M6-473 | P-0027<br>454-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428693<br>38       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven ger r<br>ece ptor<br>cyst eine<br>-rich dom<br>ain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 2-10): c.126+<br>708:TMPRSS2_c.18+8974:<br>ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}   | NA                           |
| EVT-P-002<br>2611-T01-I<br>M6-474 | P-0022<br>611-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 428736<br>29       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven ger r<br>ece ptor<br>cyst eine<br>-rich dom<br>ain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion<br>(TMPRSS2 exon 1 fused to<br>ERG exons 2-10): c.56-351<br>3:TMPRSS2_c.18+11133:ER<br>Gdel   Protein Fusion: out of<br>frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>2700-T01-I<br>M6-475 | P-0022<br>700-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 428694<br>15       | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Sca<br>ven ger r<br>ece ptor<br>cyst eine<br>-rich dom<br>ain |                   | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion<br>(TMPRSS2 exons 1-2 fused<br>to ERG exons 4-11):c.126+6<br>31:TMPRSS2_c.40-60817:ER<br>Gdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>7453-T01-I<br>M6-476 | P-0027<br>453-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872283           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11):c.56-2167:TMPRSS2_c.39+56989:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}            | NA                           |
| EVT-P-002<br>2782-T01-I<br>M6-477 | P-0022<br>782-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874883           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused in-frame with ERG exons 2-10): c.56-4767:TMPRSS2_c.18+9054:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>2865-T01-I<br>M6-478 | P-0022<br>865-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | unknown      | five_prime  | 42873095           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) ERG (NM_182918) Fusion (TMPRSS2 exons 1-2 fused with ERG exons 2-10): c.56-2979:TMPRSS2_c.90:ERGdel   Protein Fusion: mid-exon {TMPRSS2:ERG}                  | NA                           |
| EVT-P-002<br>7272-T01-I<br>M6-479 | P-0027<br>272-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872586           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-2470_c.19-26014del   Protein Fusion: out of frame {TMPRSS2:ERG}                                                                | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0023067-T01-IM6-480 | P-0023067-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42877648           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10); c.55+2229:TMPRSS2_c.18+4015:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}   | NA                           |
| EVT-P-0023270-T01-IM6-483 | P-0023270-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42874061           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon1 fused with exon4) : c.56-3945:TMPRSS2_c.40-62862:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}         | NA                           |
| EVT-P-0027210-T01-IM6-484 | P-0027210-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42870803           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused with ERG exons 4-11); c.56-687:TMPRSS2_c.39+51208:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-0023404-T01-IM6-485 | P-0023404-T01-IM6 |            | ERG-TMPRSS2 | ERG          | out-of-frame | five_prime  | 42873536           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11); c.56-3420:TMPRSS2_c.40-59054:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>6808-T01-I<br>M6-487 | P-0026<br>808-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873838           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10); c.56-3722:TMPRSS2_c.19-15805:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>3605-T01-I<br>M6-488 | P-0023<br>605-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42868088           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exons 1-2 fused with ERG exons 4-10); c.127-1583:TMPRSS2_c.-3842:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>3621-T01-I<br>M6-489 | P-0023<br>621-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869564           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused to ERG exons 2-10); c.126+482:TMPRSS2_c.19-21174:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>3625-T01-I<br>M6-490 | P-0023<br>625-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42870536           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) rearrangement: c.-56-420:TMPRSS2_chr21:g.39863632del   Protein Fusion: out of frame {TMPRSS2:ERG}                                                   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>3663-T01-I<br>M6-491 | P-0023<br>663-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42869591           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused with ERG exons 3-10); c.126+455:TMPRSS2_c.19-16612:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                           |
| EVT-P-002<br>3707-T01-I<br>M6-492 | P-0023<br>707-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867991           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) Fusion (TMPRSS2 exon2 with ERG exon 4):c.127-1486:TMPRSS2_c.40-60882:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}               | NA                           |
| EVT-P-002<br>3753-T01-I<br>M6-493 | P-0023<br>753-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42872201           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused to ERG exons 2-10); c.56-2085:TMPRSS2_c.18+21555:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}  | NA                           |
| EVT-P-002<br>3838-T01-I<br>M6-494 | P-0023<br>838-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | five_prime  | 42867929           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exons 1-2 fused to ERG exons 4-11);c.127-1424:TMPRSS2_c.40-59243:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>3839-T02-I<br>M6-495 | P-0023<br>839-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42873109           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1 fused with ERG exons 2-10); c.56-2993:TMPRSS2_c.19-8540:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                            |
| EVT-P-002<br>6768-T01-I<br>M6-496 | P-0026<br>768-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42871129           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion (TMPRSS2 exon 1 fused to ERG exons 4-11);c.56-1013:TMPRSS2_c.39+54142:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}    | NA                            |
| EVT-P-002<br>3958-T02-I<br>M6-497 | P-0023<br>958-T0<br>2-IM6 |            | ERG-TM<br>PRSS2 | ERG          | in-frame     | three_prime | 42868330           | 2               | 42                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | ERG (NM_182918) - TMPRSS2 (NM_001135099) rearrangement: c.19-2153:ERG_c.126+1716:TMPRSS2dup   Protein Fusion: in frame {ERG:TMPRSS2}                                         | NA                            |
| EVT-P-002<br>6428-T01-I<br>M6-499 | P-0026<br>428-T0<br>1-IM6 |            | ERG-TM<br>PRSS2 | ERG          | out-of-frame | five_prime  | 42874436           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exon 1 fused with ERG exons 2-10); c.56-4320:TMPRSS2_c.19-19161:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG} | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name        | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>7245-T01-I<br>M6-500 | P-0017<br>245-T0<br>1-IM6 |            | ERG-TM<br>PRSS2    | ERG          | out-of-frame | five_prime  | 42876604           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) Fusion (TMPRSS2 exon1 fused with ERG exon3) : c.55+3273:TMPRSS2_c.237-9598:ERGdel   Protein Fusion: out of frame {TMPRSS2:ERG}              | NA                           |
| EVT-P-001<br>7262-T01-I<br>M6-501 | P-0017<br>262-T0<br>1-IM6 |            | ERG-TM<br>PRSS2    | ERG          | in-frame     | five_prime  | 42868297           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TMPRSS2 exons 1-2 fused in-frame with ERG exons 2-10): c.126+1749:TMPRSS2_c.18+4187:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG} | NA                           |
| EVT-P-003<br>3832-T01-I<br>M6-502 | P-0033<br>832-T0<br>1-IM6 |            | ERG-TM<br>PRSS2    | ERG          | in-frame     | five_prime  | 42868279           | 2               | 42                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.126+1767:TMPRSS2_c.39+62909:ERGdel   Protein Fusion: in frame {TMPRSS2:ERG}                                                       | NA                           |
| EVT-P-004<br>9439-T01-I<br>M6-504 | P-0049<br>439-T0<br>1-IM6 |            | KLK3-TM<br>PRSS2   | KLK3         | unknown      | three_prime | 42843855           | 10              | 355                         | False                  | disrupted     | 0.7292576419213974       | True                |                                                  | Scavenger receptor cysteine-rich domain          | Trypsin           | KLK3 (NM_001648) - TMPRSS2 (NM_001135099) rearrangement: t(19;21)(q13.3;q22.3)(chr19:g.51359165::chr21:g.42843855)   Protein Fusion: mid-exon {KLK3:TMPRSS2}                         | NA                           |
| EVT-P-004<br>9188-T02-I<br>M7-519 | P-0049<br>188-T0<br>2-IM7 |            | MAD1L1-TM<br>PRSS2 | MAD1L1       | out-of-frame | three_prime | 42874884           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | MAD1L1 (NM_001013837) - TMPRSS2 (NM_001135099) rearrangement: t(7;21)(p22.3;q22.3)(chr7:g.2217379::chr21:g.42874884)   Protein Fusion: out of frame {MAD1L1:TMPRSS2}                 | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>9188-T01-I<br>M6-520 | P-0049<br>188-T0<br>1-IM6 |            | MAD1L1-TMPRSS2 | MAD1L1       | out-of-frame | three_prime | 42874884           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | MAD1L1 (NM_001013837) - TMPRSS2 (NM_001135099) rearrangement: t(7;21)(p22.3;q22.3)(chr7:g.2217379::chr21:g.42874884)   Protein Fusion: out of frame {MAD1L1:TMPRSS2} | NA                           |
| EVT-P-001<br>9130-T01-I<br>M6-522 | P-0019<br>130-T0<br>1-IM6 |            | MRPS6-TMPRSS2  | MRPS6        | unknown      | five_prime  | 42866482           | 3               | 50                          | False                  | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) rearrangement: chr21:g.35443032_c.150inv   Protein Fusion: mid-exon {TMPRSS2:MRPS6}                                                           | NA                           |
| EVT-P-000<br>4283-T01-I<br>M5-524 | P-0004<br>283-T0<br>1-IM5 |            | MX1-TMPRSS2    | MX1          | out-of-frame | three_prime | 42874494           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | MX1 (NM_001178046) - TMPRSS2 (NM_001135099) rearrangement : c.1759-351: MX1_c.56-4378:TMPRSS2inv   Protein fusion: out of frame (MX1-TMPRSS2)                        | NA                           |
| EVT-P-003<br>0872-T01-I<br>M6-525 | P-0030<br>872-T0<br>1-IM6 |            | MX1-TMPRSS2    | MX1          | in-frame     | three_prime | 42872650           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | MX1 (NM_001178046) - TMPRSS2 (NM_001135099) rearrangement: c.1432+622: MX1_c.56-2534:TMPRSS2inv   Protein Fusion: in frame {MX1:TMPRSS2}                             | NA                           |
| EVT-P-000<br>2901-T03-I<br>M5-527 | P-0002<br>901-T0<br>3-IM5 |            | NR3C2-TMPRSS2  | NR3C2        | out-of-frame | three_prime | 42869155           | 2               | 42                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | NR3C2-TMPRSS2 fusion : t(4;21 (q22.3;q31.23 (chr4:g.149213311::chr21:g.42869155)   Protein fusion: out of frame (NR3C2-TMPRSS2)                                      | NA                           |
| EVT-P-004<br>4218-T01-I<br>M6-530 | P-0044<br>218-T0<br>1-IM6 |            | OSBPL2-TMPRSS2 | OSBPL2       | in-frame     | three_prime | 42872237           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | OSBPL2 (NM_144498) - TMPRSS2 (NM_001135099) rearrangement: t(20;21)(q13.33;q22.3)(chr20:g.60866051::chr21:g.42872237)   Protein Fusion: in frame {OSBPL2:TMPRSS2}    | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name        | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                    | lost_domains                                     | disrupted_domains                       | source_annotation_text                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------------------------------|--------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>4876-T01-I<br>M6-531 | P-0034<br>876-T0<br>1-IM6 |            | PAXBP1-<br>TMPRSS2 | PAXBP1       | out-of-frame | three_prime | 42867134           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | PAXBP1 (NM_016631) - TMPRSS2 (NM_001135099) rearrangement: c.649+322: PAXBP1_c.127-629:TMPRSS2dup   Protein Fusion: out of frame {PAXBP1:TMPRSS2}                      | NA                           |
| EVT-P-003<br>7608-T01-I<br>M6-532 | P-0037<br>608-T0<br>1-IM6 |            | PGM5-T<br>MPRSS2   | PGM5         | in-frame     | three_prime | 42869299           | 2               | 42                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | PGM5 (NM_021965) - TMPRSS2 (NM_001135099) rearrangement: t(9;21)(q21.11;q22.3)(chr9:g.71138173::chr21:g.42869299)   Protein Fusion: in frame {PGM5:TMPRSS2}            | NA                           |
| EVT-P-003<br>0872-T01-I<br>M6-533 | P-0030<br>872-T0<br>1-IM6 |            | POLDIP3-TMPRSS2    | POLDIP3      | out-of-frame | five_prime  | 42867864           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                     | Trypsin; Scavenger receptor cysteine-rich domain |                                         | TMPRSS2 (NM_001135099) - POLDIP3 (NM_032311) rearrangement: t(21;22)(q22.3;q13.2)(chr21:g.42867864::chr22:g.43009989)   Protein Fusion: out of frame {TMPRSS2:POLDIP3} | NA                           |
| EVT-P-006<br>3301-T01-I<br>M7-534 | P-0063<br>301-T0<br>1-IM7 |            | PREX2-TMPRSS2      | PREX2        | unknown      | three_prime | 42852468           | 6               | 206                         | False                  | retained      | 1.0                      | False               | Trypsin                                             |                                                  | Scavenger receptor cysteine-rich domain | PREX2 (NM_024870) - TMPRSS2 (NM_001135099) rearrangement: t(8;21)(q13.2;q22.3)(chr8:g.68911596::chr21:g.42852468)   Protein Fusion: mid-exon {PREX2:TMPRSS2}           | NA                           |
| EVT-P-004<br>6175-T02-I<br>M6-535 | P-0046<br>175-T0<br>2-IM6 |            | RCN1-TMPRSS2       | RCN1         | out-of-frame | three_prime | 42875275           | 1               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | RCN1 (NM_002901) - TMPRSS2 (NM_001135099) rearrangement: t(11;21)(p13;q22.3)(chr11:g.32114150::chr21:g.42875275)   Protein Fusion: out of frame {RCN1:TMPRSS2}         | NA                           |
| EVT-P-000<br>6603-T01-I<br>M5-536 | P-0006<br>603-T0<br>1-IM5 |            | SETD4-TMPRSS2      | SETD4        | out-of-frame | three_prime | 42869417           | 2               | 42                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | SETD4 (NM_017438) - TMPRSS2 (NM_001135099) rearrangement: c.169+807: SETD4_c.126+629:TMPRSS2del   Protein fusion: out of frame (SETD4-TMPRSS2)                         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                    | lost_domains                                     | disrupted_domains                       | source_annotation_text                                                                                                                                          | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------------------------------|--------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>9843-T01-I<br>M6-537 | P-0029<br>843-T0<br>1-IM6 |            | SH3BGR-TMPRSS2 | SH3BGR       | out-of-frame | three_prime | 42872212           | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | SH3BGR (NM_007341) - TMPRSS2 (NM_001135099) rearrangement: c.421-1136:SH3BGR_c.56-2096:TMPRSS2inv   Protein Fusion: out of frame {SH3BGR:TMPRSS2}               | NA                           |
| EVT-P-001<br>8122-T01-I<br>M6-539 | P-0018<br>122-T0<br>1-IM6 |            | TMPRSS2-ABCG1  | ABCG1        | out-of-frame | five_prime  | 42851084           | 7               | 265                         | True                   | lost          | 0.0                      | False               |                                                     | Trypsin                                          | Scavenger receptor cysteine-rich domain | TMPRSS2 (NM_001135099) - ABCG1 (NM_207627) Rearrangement: c.794+15:TMPRSS2_c.48+6119:ABCG1inv   Protein Fusion: out of frame {TMPRSS2:ABCG1}                    | NA                           |
| EVT-P-000<br>2123-T02-I<br>M5-540 | P-0002<br>123-T0<br>2-IM5 |            | TMPRSS2-ACP3   | ACP3         | unknown      | five_prime  | 42868047           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                     | Trypsin; Scavenger receptor cysteine-rich domain |                                         | TMPRSS2 (NM_001135099) - ACP3 (NM_001134194) rearrangement: t(3;21)(q22.1;q22.3)(chr3:g.132053847::chr21:g.42868047)   Antisense fusion                         | NA                           |
| EVT-P-000<br>6689-T01-I<br>M5-541 | P-0006<br>689-T0<br>1-IM5 |            | TMPRSS2-AFDN   | AFDN         | in-frame     | five_prime  | 42875775           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                                                     | Trypsin; Scavenger receptor cysteine-rich domain |                                         | TMPRSS2 (NM_001135099) - MLLT4 (NM_001207008) rearrangement : t(6;21)(q27;q22.3)(chr6:168293792::chr21:42875775)   Protein fusion: in frame (TMPRSS2-MLLT4)     | NA                           |
| EVT-P-000<br>2398-T01-I<br>M3-542 | P-0002<br>398-T0<br>1-IM3 |            | TMPRSS2-ARF4   | ARF4         | out-of-frame | three_prime | 42867618           | 3               | 43                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor or cysteine-rich domain |                                                  |                                         | ARF4 (NM_001660) - TMPRSS2 (NM_001135099) Translocation : t(3;21)(p14.3;q22.3)(chr3:g.57572612::chr21:g.42867618)   Protein fusion: out of frame (ARF4-TMPRSS2) | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4395-T01-I<br>M6-543 | P-0014<br>395-T0<br>1-IM6 |            | TPRSS2-ARHGAP26 | ARHGAP26     | in-frame     | five_prime  | 42868024           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ARHGAP26 (NM_015071) fusion (TPRSS2 exons 1-2 fused in-frame to ARHGAP26 exons 12-23): t(5;21)(q31.3;q22.3)(chr5:g.142324441::chr21:g.42868024)   Protein fusion: in frame (TPRSS2-ARHGAP26) | NA                           |
| EVT-P-006<br>4697-T01-I<br>M7-545 | P-0064<br>697-T0<br>1-IM7 |            | TPRSS2-ARRB1    | ARRB1        | out-of-frame | five_prime  | 42877301           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ARRB1 (NM_004041) rearrangement: t(11;21)(q13.4;q22.3)(chr11:g.75013239::chr21:g.42877301)   Protein Fusion: out of frame {TPRSS2:ARRB1}                                                     | NA                           |
| EVT-P-005<br>1214-T01-I<br>M6-546 | P-0051<br>214-T0<br>1-IM6 |            | TPRSS2-BRAF     | BRAF         | in-frame     | five_prime  | 42867846           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846)   Protein Fusion: in frame {TPRSS2:BRAF}                                                                     | NA                           |
| EVT-P-001<br>8524-T01-I<br>M6-547 | P-0018<br>524-T0<br>1-IM6 |            | TPRSS2-BRAF     | BRAF         | out-of-frame | five_prime  | 42873801           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337)   Protein Fusion: out of frame {TPRSS2:BRAF}                                                         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                | disrupted_domains | source_annotation_text                                                                                                                                                            | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|-------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>2778-T01-I<br>M6-549 | P-0022<br>778-T0<br>1-IM6 |            | TPRSS<br>S2-BRAF  | BRAF         | in-frame     | five_prime  | 42878796           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TPRSS2<br>(NM_001135099) - BRAF<br>(NM_004333)<br>Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796)   Protein<br>Fusion: in frame<br>{TPRSS2:BRAF}           | NA                           |
| EVT-P-001<br>3729-T01-I<br>M5-551 | P-0013<br>729-T0<br>1-IM5 |            | TPRSS<br>S2-CASZ1 | CASZ1        | out-of-frame | five_prime  | 42868002           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TPRSS2<br>(NM_001135099) - CASZ1<br>(NM_001079843)<br>rearrangement: t(1;21)(p36.22;q22.3)(chr1:g.10703925::chr21:g.42868002)   Protein<br>fusion: out of frame<br>(TPRSS2-CASZ1) | NA                           |
| EVT-P-002<br>6514-T01-I<br>M6-552 | P-0026<br>514-T0<br>1-IM6 |            | TPRSS<br>S2-CCN6  | CCN6         | out-of-frame | five_prime  | 42879671           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TPRSS2<br>(NM_001135099) - WISP3<br>(NM_003880)<br>Rearrangement : t(6;21)(q21;q22.2)(chr6:g.112390072::chr21:g.30301865)   Protein<br>Fusion: out of frame<br>{TPRSS2:WISP3}     | NA                           |
| EVT-P-002<br>2953-T01-I<br>M6-553 | P-0022<br>953-T0<br>1-IM6 |            | TPRSS<br>S2-CERS6 | CERS6        | out-of-frame | five_prime  | 42873665           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin;<br>Scavenge<br>receptor<br>cysteine-rich<br>domain |                   | TPRSS2<br>(NM_001135099) - CERS6<br>(NM_203463)<br>rearrangement: t(2;21)(q24.3;q22.3)(chr2:g.169560626::chr21:g.42873665)   Protein<br>Fusion: out of frame<br>{TPRSS2:CERS6}    | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name           | partner_gene   | frame_status | target_role             | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                            | lost_domains                                                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                   | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------------|----------------|--------------|-------------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>7882-T02-I<br>M6-554 | P-0017<br>882-T0<br>2-IM6 |            | TPRS<br>S2-CHD<br>3   | CH<br>D3       | unknown      | fiv<br>e_<br>pri<br>me  | 428704<br>83       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                                             | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                   | TPRSS2<br>(NM_001135099)<br>rearrangement: t(17;21)(p13<br>.1;q22.3)(chr17:g.7790479::<br>chr21:g.42870483)  <br>Antisense Fusion                                                        | NA                           |
| EVT-P-000<br>2971-T02-I<br>M5-555 | P-0002<br>971-T0<br>2-IM5 |            | TPRS<br>S2-CHD<br>3   | CH<br>D3       | in-frame     | thr<br>ee_<br>pr<br>ime | 428716<br>18       | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsi<br>n; Sca<br>venger<br>recept<br>or cyst<br>eine-ri<br>ch do<br>main |                                                                                              |                   | CHD3 (NM_001005271) -<br>TPRSS2<br>(NM_001135099)<br>Rearrangement : t(17;21)(17<br>p13.1;21q22.3)(chr17:g.781<br>1475::chr21:g.42871618)  <br>Protein fusion: in frame<br>(CHD3-TPRSS2) | NA                           |
| EVT-P-001<br>7882-T01-I<br>M6-556 | P-0017<br>882-T0<br>1-IM6 |            | TPRS<br>S2-CHD<br>3   | CH<br>D3       | unknown      | fiv<br>e_<br>pri<br>me  | 428704<br>83       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                                             | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                   | TPRSS2<br>(NM_001135099)<br>rearrangement: t(17;21)(p13<br>.1;q22.3)(chr17:g.7790479::<br>chr21:g.42870483)  <br>Antisense Fusion                                                        | NA                           |
| EVT-P-001<br>6386-T01-I<br>M6-557 | P-0016<br>386-T0<br>1-IM6 |            | TPRS<br>S2-CHS<br>T12 | CH<br>ST1<br>2 | unknown      | fiv<br>e_<br>pri<br>me  | 428724<br>53       | 2               | 19                          | True                   | lost          | 0.0                      | False               |                                                                             | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                   | TPRSS2<br>(NM_001135099)<br>rearrangement: t(7;21)(p22.<br>3;q22.3)(chr7:g.2452854::ch<br>r21:g.42872453)   Transcript<br>Fusion<br>{TPRSS2:CHST12}                                      | NA                           |
| EVT-P-003<br>4620-T01-I<br>M6-558 | P-0034<br>620-T0<br>1-IM6 |            | TPRS<br>S2-CSK        | CSK            | unknown      | thr<br>ee_<br>pr<br>ime | 428758<br>44       | 1               | 19                          | True                   | retained      | 1.0                      | False               | Trypsi<br>n; Sca<br>venger<br>recept<br>or cyst<br>eine-ri<br>ch do<br>main |                                                                                              |                   | CSK (NM_001127190) -<br>TPRSS2<br>(NM_001135099)<br>rearrangement: t(15;21)(q24<br>.1;q22.3)(chr15:g.75094115:<br>:chr21:g.42875844)   Protein<br>Fusion: mid-exon<br>{CSK:TPRSS2}       | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name            | partner_gene    | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|------------------------|-----------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|--------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>3701-T02-I<br>M6-559 | P-0033<br>701-T0<br>2-IM6 |            | TPRS<br>S2-CYP<br>3A43 | CYP<br>3A4<br>3 | in-frame     | three_prime | 428738<br>43       | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | CYP3A43 (NM_022820) - TMPRSS2 (NM_001135099) rearrangement: t(7;21)(q22.1;q22.3)(chr7:g.99448887::chr21:g.42873843)   Protein Fusion: in frame {CYP3A43:TMPRSS2}     | NA                           |
| EVT-P-000<br>5047-T01-I<br>M5-560 | P-0005<br>047-T0<br>1-IM5 |            | TPRS<br>S2-CYP<br>4Z1  | CYP<br>4Z1      | out-of-frame | three_prime | 428747<br>35       | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | CYP4Z1 (NM_178134) - TMPRSS2 (NM_001135099) rearrangement : t(1;21)(1p33;21q22.3)(chr1:g.47561132::chr21:g.42874735)   Protein fusion: out of frame (CYP4Z1-TMPRSS2) | NA                           |
| EVT-P-001<br>2695-T01-I<br>M5-561 | P-0012<br>695-T0<br>1-IM5 |            | TPRS<br>S2-DEN<br>ND3  | DEN<br>ND3      | out-of-frame | three_prime | 428738<br>72       | 2               | 19                          | True                   | retained      | 1.0                      | False               | Trypsin; Scavenger receptor cysteine-rich domain |                                                  |                   | DENND3 - TMPRSS2 rearrangement: t(8;21)(q24.3;q22.3)(chr8:g.142204112::chr21:g.42873872)   Protein fusion: out of frame (DENND3-TMPRSS2)                             | NA                           |
| EVT-P-000<br>5459-T02-I<br>M6-562 | P-0005<br>459-T0<br>2-IM6 |            | TPRS<br>S2-DSC<br>AM   | DSC<br>AM       | out-of-frame | five_prime  | 428679<br>10       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - DSCAM (NM_001389) rearrangement: c.127-1405:TMPRSS2_c.3562+3890:DSCAMdel   Protein Fusion: out of frame {TMPRSS2:DSCAM}                     | NA                           |
| EVT-P-005<br>9208-T01-I<br>M7-564 | P-0059<br>208-T0<br>1-IM7 |            | TPRS<br>S2-DYR<br>K1A  | DYR<br>K1A      | in-frame     | five_prime  | 428679<br>55       | 3               | 43                          | True                   | lost          | 0.0                      | False               |                                                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TMPRSS2 (NM_001135099) - DYRK1A (NM_001396) rearrangement: c.127-1450:TMPRSS2_c.1240-2605:DYRK1Ainv   Protein Fusion: in frame {TMPRSS2:DYRK1A}                      | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                             | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>9689-T01-I<br>M6-566 | P-0039<br>689-T0<br>1-IM6 |            | TPRSS2-ERG  | ERG          | out-of-frame | five_prime  | 42870319           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-203:TPRSS2_c.19-1128:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG}       | NA                           |
| EVT-P-005<br>1162-T01-I<br>M6-567 | P-0051<br>162-T0<br>1-IM6 |            | TPRSS2-ERG  | ERG          | in-frame     | five_prime  | 42868272           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) rearrangement: c.127-1767:TPRSS2_c.19-18536:ERGdel   Protein Fusion: in frame {TPRSS2:ERG} | NA                           |
| EVT-P-003<br>4813-T03-I<br>M6-568 | P-0034<br>813-T0<br>3-IM6 |            | TPRSS2-ERG  | ERG          | out-of-frame | five_prime  | 42876420           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+3457:TPRSS2_c.18+4434:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG}      | NA                           |
| EVT-P-004<br>0754-T01-I<br>M6-569 | P-0040<br>754-T0<br>1-IM6 |            | TPRSS2-ERG  | ERG          | unknown      | five_prime  | 42870317           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.56-201:TPRSS2_c.40-54954:ERGdel   Transcript Fusion {TPRSS2:ERG}                 | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                             | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>8200-T01-I<br>M6-570 | P-0058<br>200-T0<br>1-IM6 |            | TPRSS<br>S2-ERG | ERG          | out-of-frame | five_prime  | 42870336           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.56-220:TPRSS2_c.18+15563:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG}      | NA                           |
| EVT-P-006<br>7541-T02-I<br>M7-571 | P-0067<br>541-T0<br>2-IM7 |            | TPRSS<br>S2-ERG | ERG          | out-of-frame | five_prime  | 42877906           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+1971:TPRSS2_c.18+22187:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG}     | NA                           |
| EVT-P-005<br>4566-T02-I<br>M6-572 | P-0054<br>566-T0<br>2-IM6 |            | TPRSS<br>S2-ERG | ERG          | in-frame     | five_prime  | 42867035           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_004449) fusion: c.127-530:TPRSS2_c.40-59799:ERGdel   Protein Fusion: in frame {TPRSS2:ERG}         | NA                           |
| EVT-P-005<br>4527-T01-I<br>M6-573 | P-0054<br>527-T0<br>1-IM6 |            | TPRSS<br>S2-ERG | ERG          | in-frame     | five_prime  | 42868216           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) rearrangement: c.127-1711:TPRSS2_c.19-24889:ERGdel   Protein Fusion: in frame {TPRSS2:ERG} | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                                | disrupt_ed_do_mains | source_annotation_text                                                                                                                                | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-006<br>7290-T01-I<br>M7-574 | P-0067<br>290-T0<br>1-IM7 |              | TMPRS<br>S2-ERG | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428731<br>35          | 2                 | 19                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.56-3<br>019:TMPRSS2_c.39+40804<br>:ERGdel   Protein Fusion:<br>out of frame<br>{TMPRSS2:ERG} | NA                            |
| EVT-P-006<br>6853-T01-I<br>M7-576 | P-0066<br>853-T0<br>1-IM7 |              | TMPRS<br>S2-ERG | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428685<br>19          | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.126+<br>1527:TMPRSS2_c.18+4887<br>:ERGdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}        | NA                            |
| EVT-P-006<br>6769-T01-I<br>M7-577 | P-0066<br>769-T0<br>1-IM7 |              | TMPRS<br>S2-ERG | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428699<br>29          | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) -ERG<br>()rearrangement: c.126+117<br>:TMPRSS2_c.40-57447:ER<br>Gdel   Protein Fusion: in<br>frame {TMPRSS2:ERG}            | NA                            |
| EVT-P-006<br>6737-T01-I<br>M7-578 | P-0066<br>737-T0<br>1-IM7 |              | TMPRS<br>S2-ERG | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428686<br>36          | 2                 | 42                           | True                     | lost           | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TMPRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.126+<br>1410:TMPRSS2_c.39+6165<br>6:ERGdel   Protein Fusion:<br>in frame {TMPRSS2:ERG}       | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name     | part_ner_gene | fra_me_status        | tar_get_role           | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_doma_in | lost_doma_in                                                                                 | disrupt_ed_doma_in | source_annotation_text                                                                                                                                                                   | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------|---------------|----------------------|------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-------------------|----------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>8080-T01-I<br>M6-579 | P-0028<br>080-T0<br>1-IM6 |              | TPPRS<br>S2-ERG | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428750<br>20          | 1                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TPMRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.55+4<br>857:TPMRSS2_c.18+9341:<br>ERGdel   Protein Fusion: out<br>of frame {TPMRSS2:ERG}                                        | NA                            |
| EVT-P-006<br>6733-T01-I<br>M7-580 | P-0066<br>733-T0<br>1-IM7 |              | TPPRS<br>S2-ERG | ER<br>G       | in-fr<br>ame         | fiv<br>e_<br>pri<br>me | 428683<br>87          | 2                | 42                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TPMRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.126+<br>1659:TPMRSS2_c.18+9645<br>:ERGdel   Protein Fusion: in<br>frame {TPMRSS2:ERG}                                           | NA                            |
| EVT-P-002<br>2117-T04-I<br>M7-581 | P-0022<br>117-T0<br>4-IM7 |              | TPPRS<br>S2-ERG | ER<br>G       | unk<br>now<br>n      | fiv<br>e_<br>pri<br>me | 428663<br>28          | 3                | 102                          | False                    | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TPMRSS2<br>(NM_001135099) - ERG<br>(NM_182918) fusion: c.304:<br>TPMRSS2_c.18+8044:ERG<br>del   Protein Fusion:<br>mid-exon {TPMRSS2:ERG}                                                | NA                            |
| EVT-P-002<br>1782-T01-I<br>M6-582 | P-0021<br>782-T0<br>1-IM6 |              | TPPRS<br>S2-ERG | ER<br>G       | out-<br>of-fr<br>ame | fiv<br>e_<br>pri<br>me | 428728<br>68          | 2                | 19                           | True                     | lost           | 0.0                        | False                |                   | Tryp<br>sin;<br>Sca<br>ven<br>ger<br>r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                    | TPMRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TPMRSS2 exon 1 with<br>ERG exon4) : c.56-2752:TM<br>PRSS2_c.39+30375:ERGde<br>l   Protein Fusion: out of<br>frame {TPMRSS2:ERG} | NA                            |

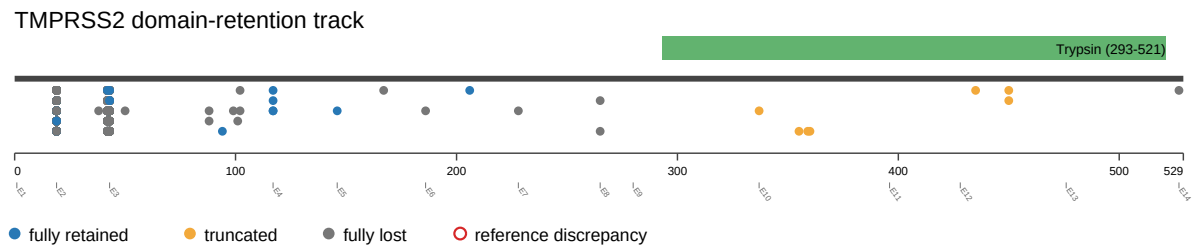
| event_id                          | sampl_e_id                | pa_tie_nt_i_d | fusion_name     | part_ner_ge_ne | fra_me_st_atu_s  | tar_get_r_ole   | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intro_nic_bre_akpoint | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_ncated | retain_ed_do_mains | lost_do_mains                                                                             | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                        | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|---------------|-----------------|----------------|------------------|-----------------|-----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|--------------------|-------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>1561-T01-I<br>M6-584 | P-0021<br>561-T0<br>1-IM6 |               | TPPRS<br>S2-ERG | ER<br>G        | in-fr<br>ame     | fiv_e_pri<br>me | 428675<br>76          | 3                 | 43                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TPMRSS2<br>(NM_001135099) -ERG<br>(NM_004449) Fusion<br>(TPMRSS2 exon2 with ERG<br>exon4) : c.127-1071:TPPRS<br>S2_c.40-56818:ERGdel  <br>Protein Fusion: in frame<br>{TPMRSS2:ERG}           | NA                            |
| EVT-P-006<br>7735-T01-I<br>M7-585 | P-0067<br>735-T0<br>1-IM7 |               | TPPRS<br>S2-ERG | ER<br>G        | unk<br>now<br>n  | fiv_e_pri<br>me | 428682<br>18          | 3                 | 43                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TPMRSS2<br>(NM_001135099) - ERG<br>(NM_004449) fusion: c.127-<br>1713:TPMRSS2_c.40-5524<br>7:ERGdel   Transcript<br>Fusion {TPMRSS2:ERG}                                                      | NA                            |
| EVT-P-002<br>1297-T01-I<br>M6-586 | P-0021<br>297-T0<br>1-IM6 |               | TPPRS<br>S2-ERG | ER<br>G        | out-of-fr<br>ame | fiv_e_pri<br>me | 428691<br>75          | 2                 | 42                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TPMRSS2<br>(NM_001135099) - ERG<br>(NM_182918) (TPMRSS2<br>exons 1-2 fused with ERG<br>exons 3-10):c.126+871:TMP<br>RSS2_c.237-7837:ERGdel  <br>Protein Fusion: out of frame<br>{TPMRSS2:ERG} | NA                            |
| EVT-P-005<br>4521-T01-I<br>M6-587 | P-0054<br>521-T0<br>1-IM6 |               | TPPRS<br>S2-ERG | ER<br>G        | unk<br>now<br>n  | fiv_e_pri<br>me | 428738<br>49          | 2                 | 19                           | True                     | lost            | 0.0                        | False                |                    | Tryp<br>sin;<br>Sca<br>ven<br>ger r<br>ece<br>ptor<br>cyst<br>eine<br>-rich<br>dom<br>ain |                     | TPMRSS2<br>(NM_001135099) - ERG<br>(NM_004449)<br>rearrangement: c.56-3733:T<br>MPRSS2_c.40-54513:ERGd<br>el   Transcript Fusion<br>{TPMRSS2:ERG}                                             | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                     | disrupted_domains | source_annotation_text                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>4506-T01-I<br>M6-588 | P-0054<br>506-T0<br>1-IM6 |            | TPRSS2-ERG  | ERG          | out-of-frame | five_prime  | 42874701           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_004449) rearrangement: c.56-4585:TMPRSS2_c.40-59474:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG}                                 | NA                           |
| EVT-P-005<br>4972-T01-I<br>M6-589 | P-0054<br>972-T0<br>1-IM6 |            | TPRSS2-ERG  | ERG          | out-of-frame | five_prime  | 42878690           | 1               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) fusion: c.55+187:TMPRSS2_c.236+9326:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG}                                         | NA                           |
| EVT-P-002<br>1104-T02-I<br>M6-590 | P-0021<br>104-T0<br>2-IM6 |            | TPRSS2-ERG  | ERG          | unknown      | five_prime  | 42866560           | 3               | 43                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_004449) rearrangement: c.127-55:TMPRSS2_c.40-55445:ERGdel   Transcript Fusion {TPRSS2:ERG}                                             | NA                           |
| EVT-P-002<br>0984-T01-I<br>M6-591 | P-0020<br>984-T0<br>1-IM6 |            | TPRSS2-ERG  | ERG          | out-of-frame | five_prime  | 42871570           | 2               | 19                          | True                   | lost          | 0.0                      | False               |                  | Trypsin; Scavenger receptor cysteine-rich domain |                   | TPRSS2 (NM_001135099) - ERG (NM_182918) fusion (TPRSS2 exon 1 fused to ERG exons 2-10): c.56-1454:TMPRSS2_c.18+8460:ERGdel   Protein Fusion: out of frame {TPRSS2:ERG} | NA                           |

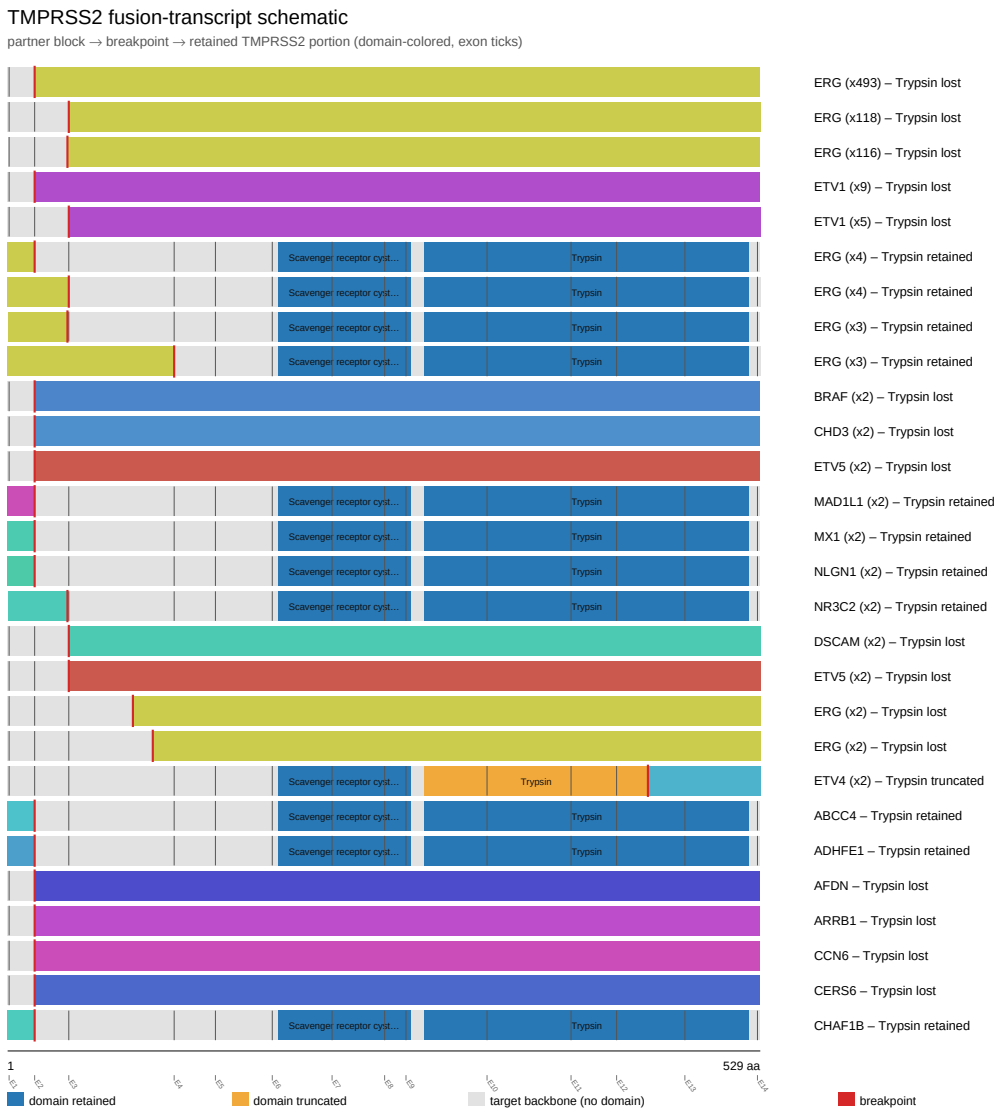
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# Figures

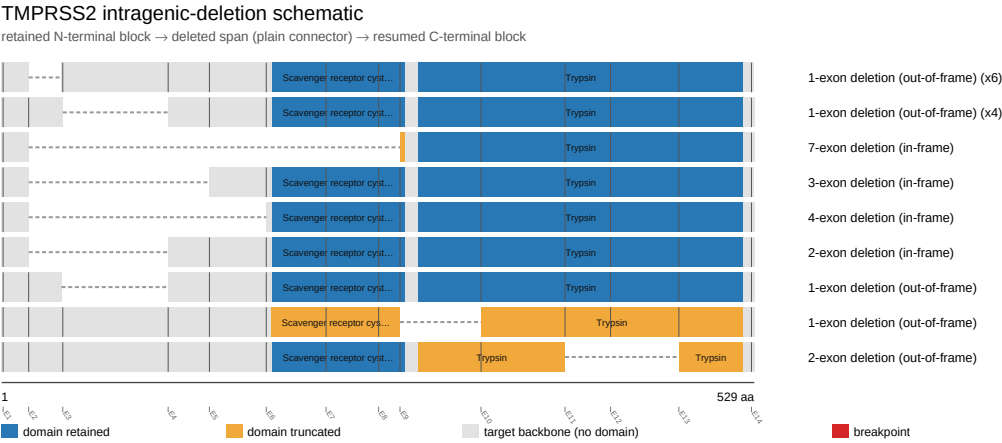
## domain retention outliers



## fusion schematic



# intragenic deletion schematic



## reference comparison

Reference vs msk\_impact\_50k\_2026

